

A Short Practical Introduction to Bayesian Analysis

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24–25 September 2026

A short worked example

The Bayesian toolkit

The Bayesian workflow

Working through the workflow: golf putting

Becoming independent Bayesian analysts

Course notebooks, slides, and data

github.com/opherdonchin/BayesShortCourse

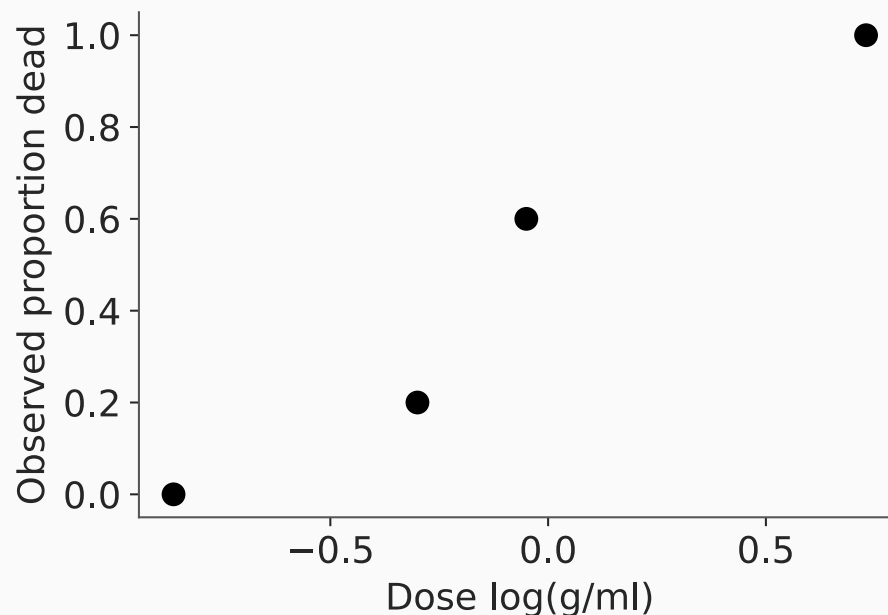
A short worked example

Dose response

Four doses, twenty animals

A toxin is given to rats at four doses, five rats per dose.

log dose	rats	deaths
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5

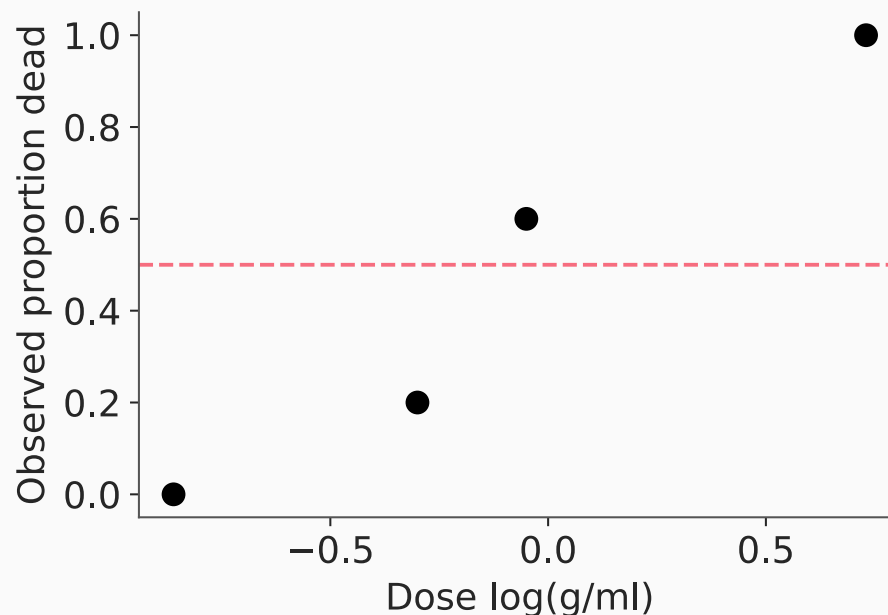


Data: Racine-Poon et al. (1986); Gelman & Vehtari, *Bayesian Workflow*, §3.5

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A generative model for mortality

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

$$\alpha \sim \text{Normal}(0, 5)$$

$$\beta \sim \text{HalfNormal}(5)$$

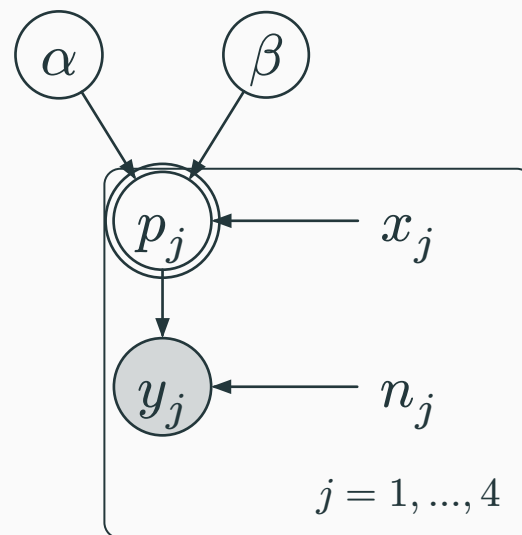
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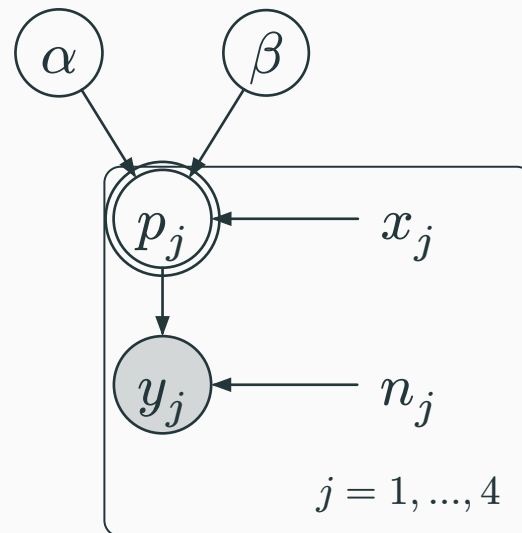
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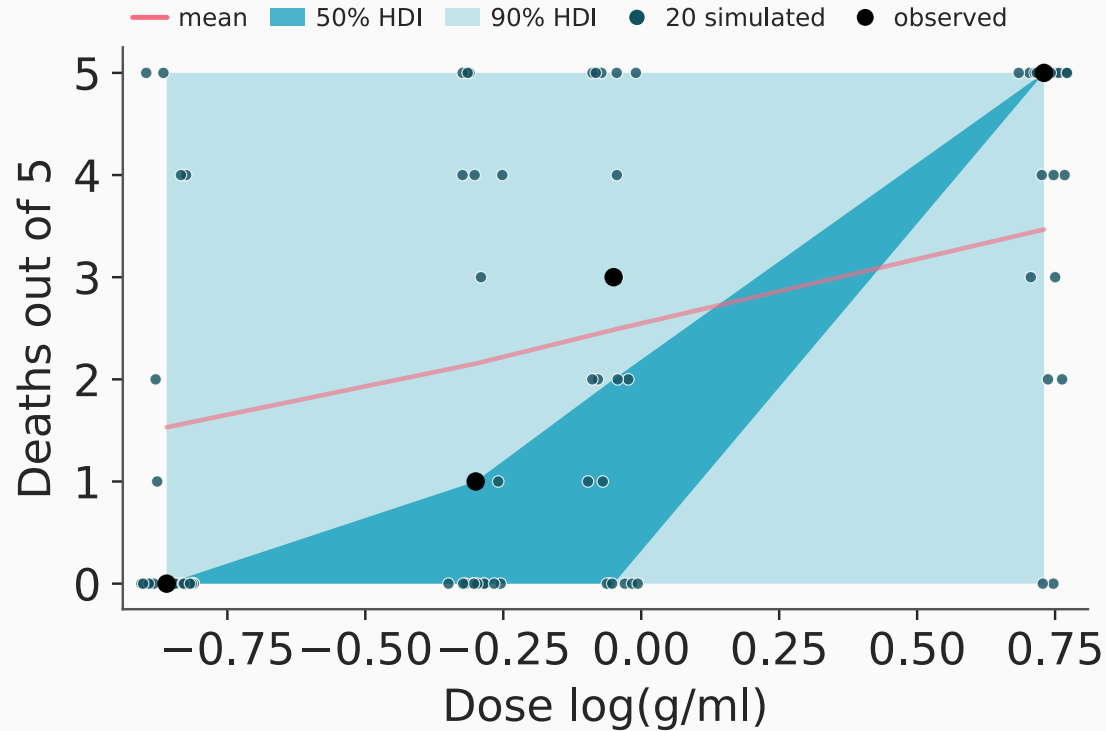
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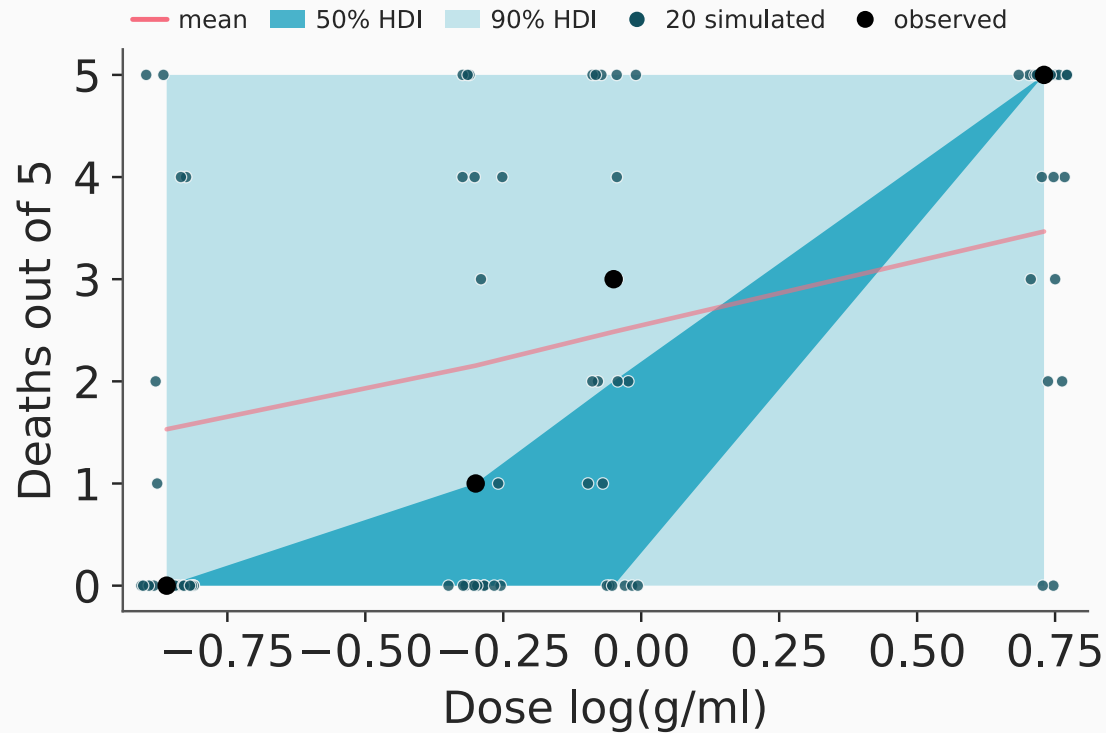
```
with pm.Model() as model:  
    alpha = pm.Normal("alpha", mu=0, sigma=5)  
    beta = pm.HalfNormal("beta", sigma=5)  
    logit_p = alpha + beta * dose  
    pm.Binomial("deaths", n=n, logit_p=logit_p, observed=deaths)
```

Priors: what could we have seen?



Prior predictive data should not

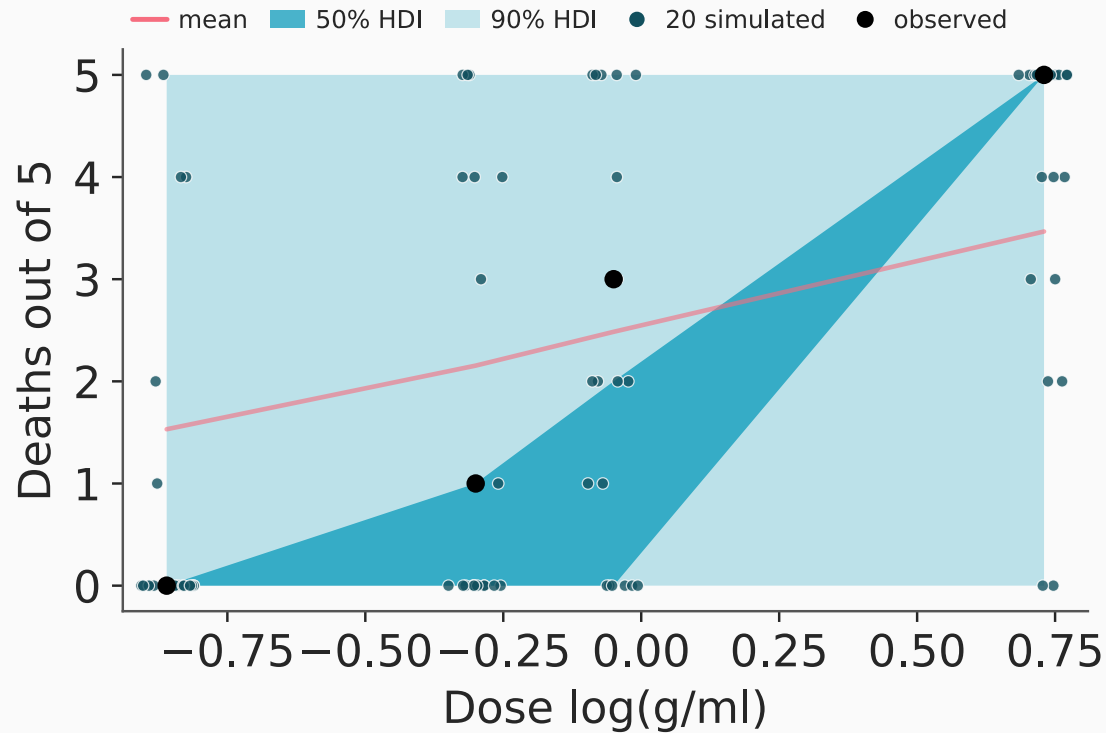
Priors: what could we have seen?



Prior predictive data should not

- violate common sense

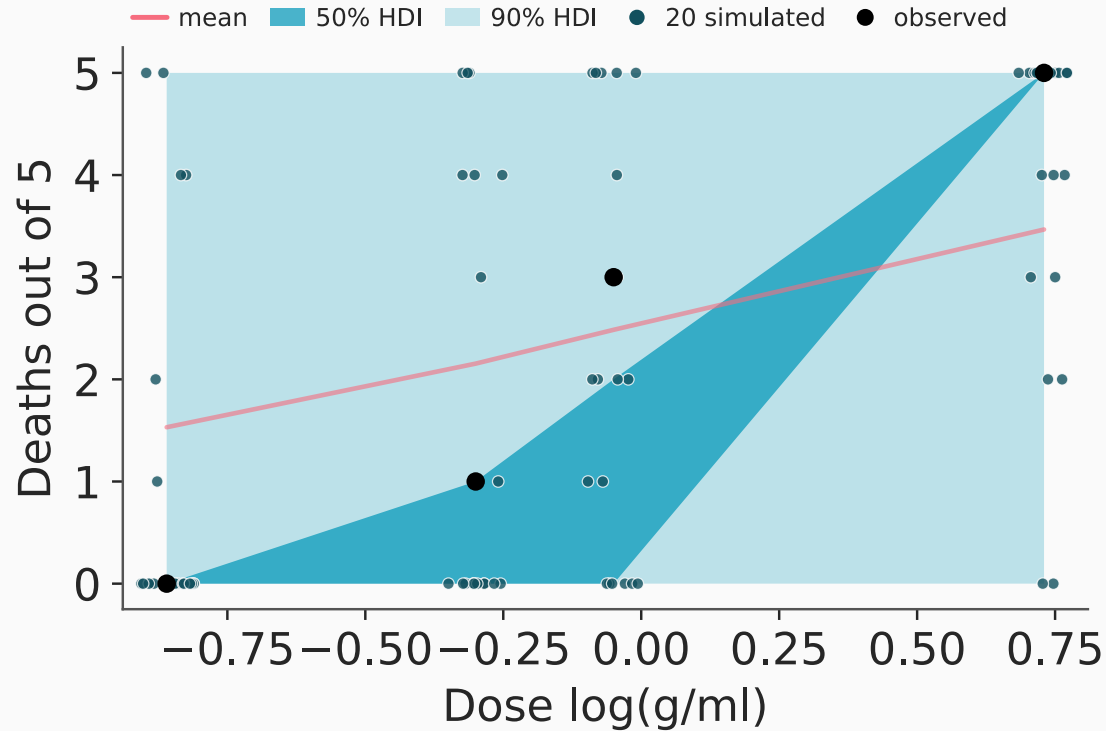
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- make our data highly unlikely

Priors: what could we have seen?



Prior predictive data should not

- violate common sense
- make our data highly unlikely
- confine the model to our data

Sampling and diagnostics

Fitting the model means drawing samples from the posterior distribution.

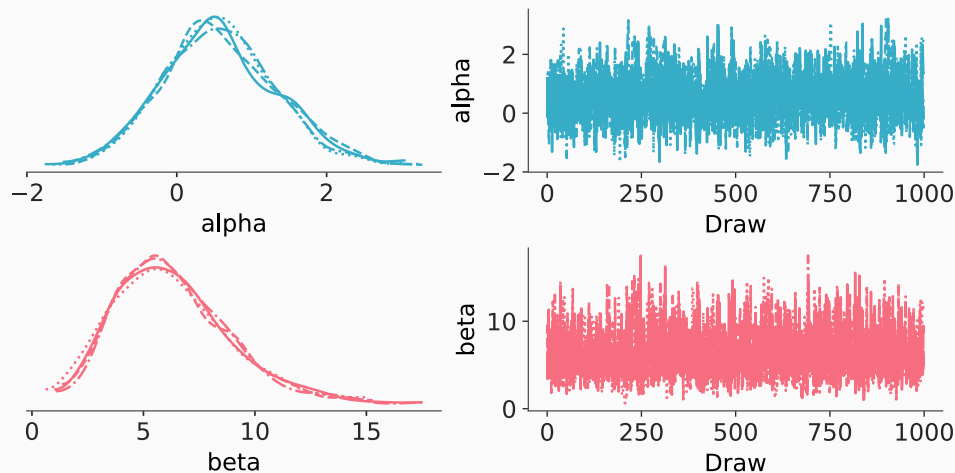
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with model:  
    idata = pm.sample(draws=1000, chains=4)
```

Sampling and diagnostics

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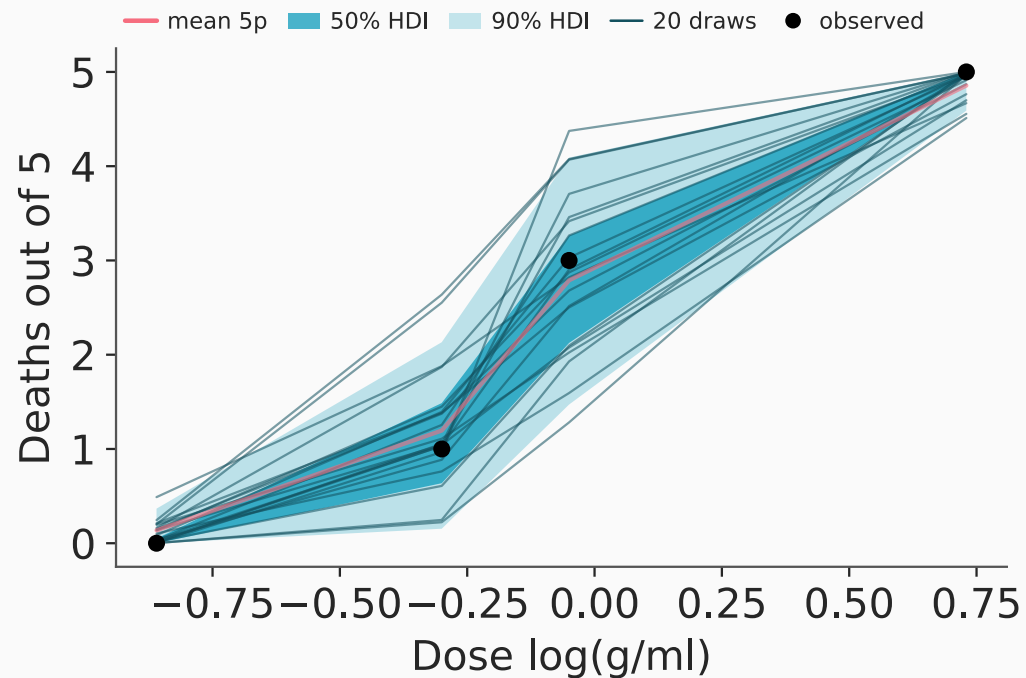


Check before interpreting:

- 0 divergences
- $\hat{R} = 1.00$
- ESS > 1,300 of 4,000 draws

Posterior: which dose-response curves remain plausible?

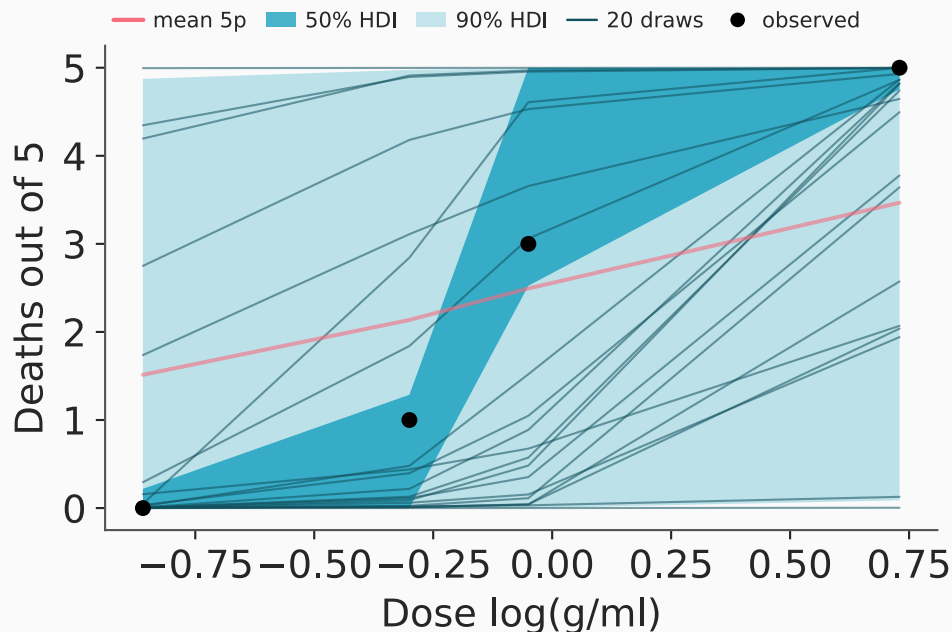
The bands show uncertainty about the mean number of deaths.



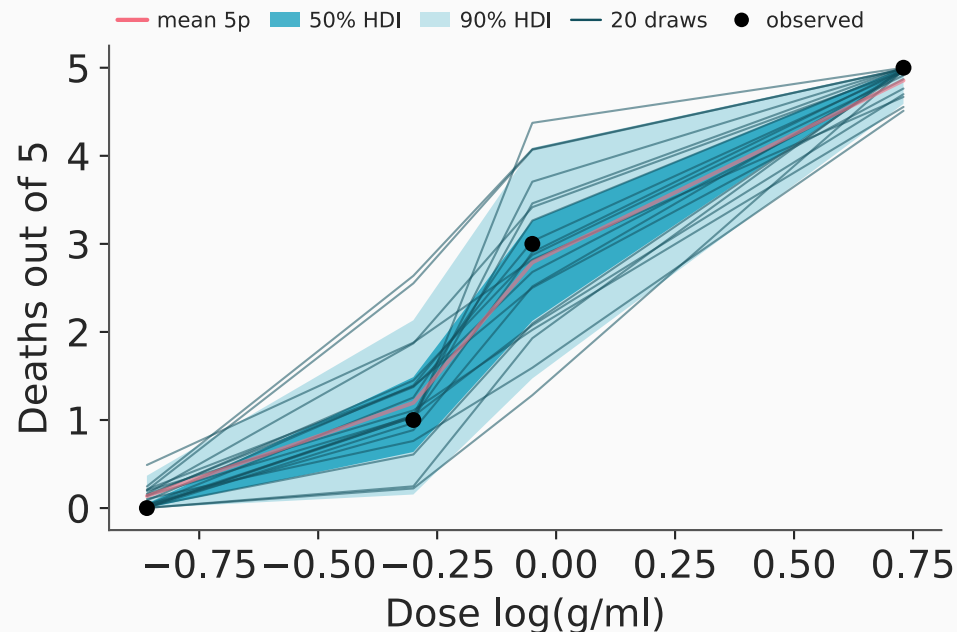
Posterior: which dose-response curves remain plausible?

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Prior

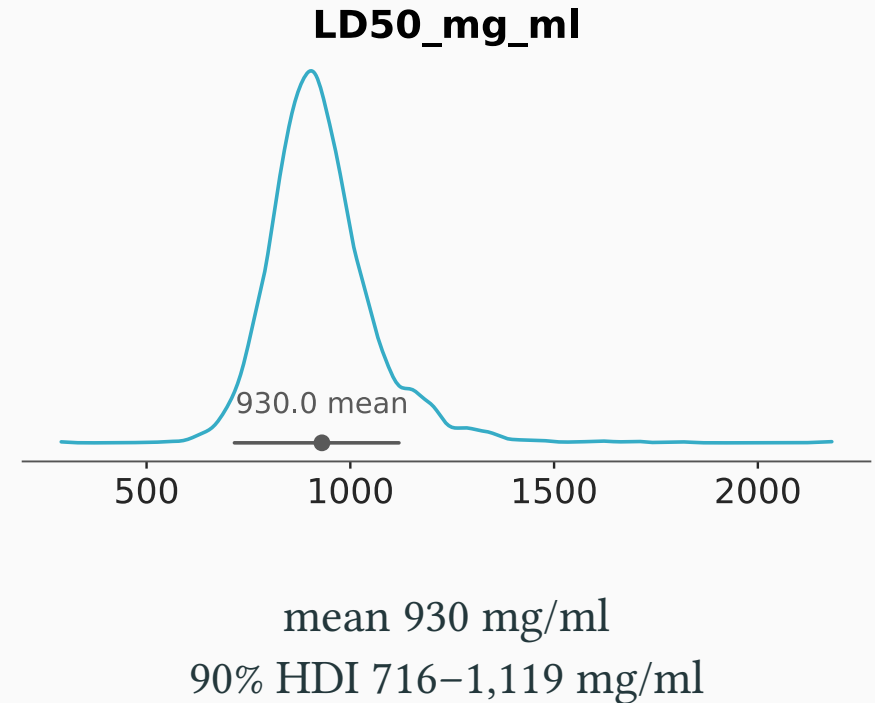


Posterior



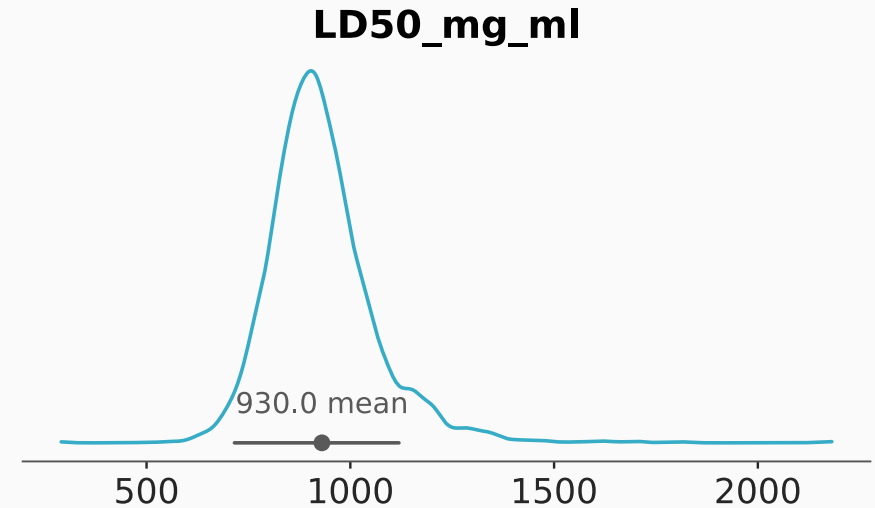
LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.



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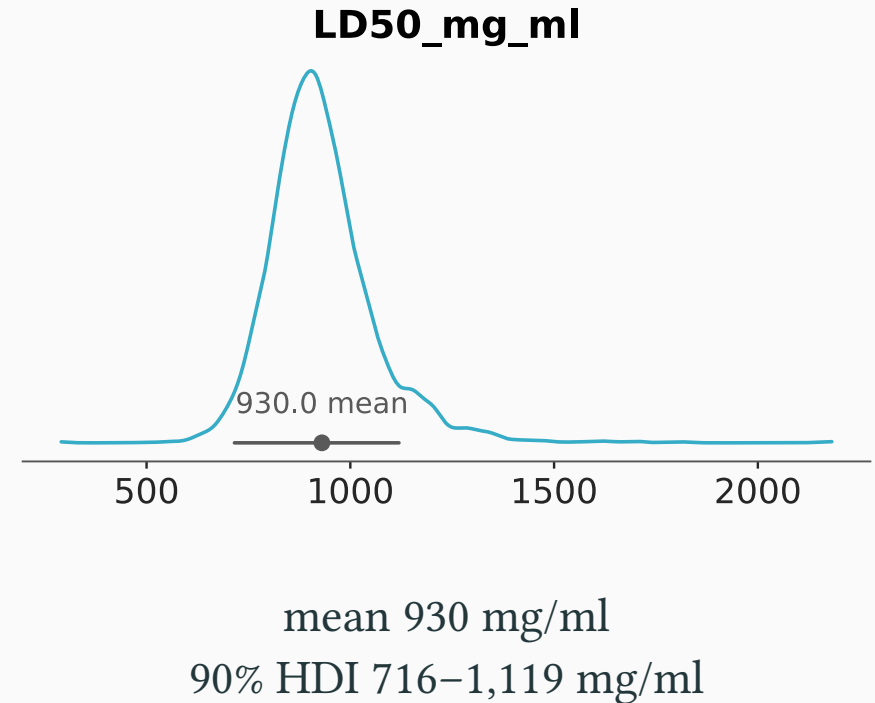
- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.
- So every quantity comes with its own uncertainty.



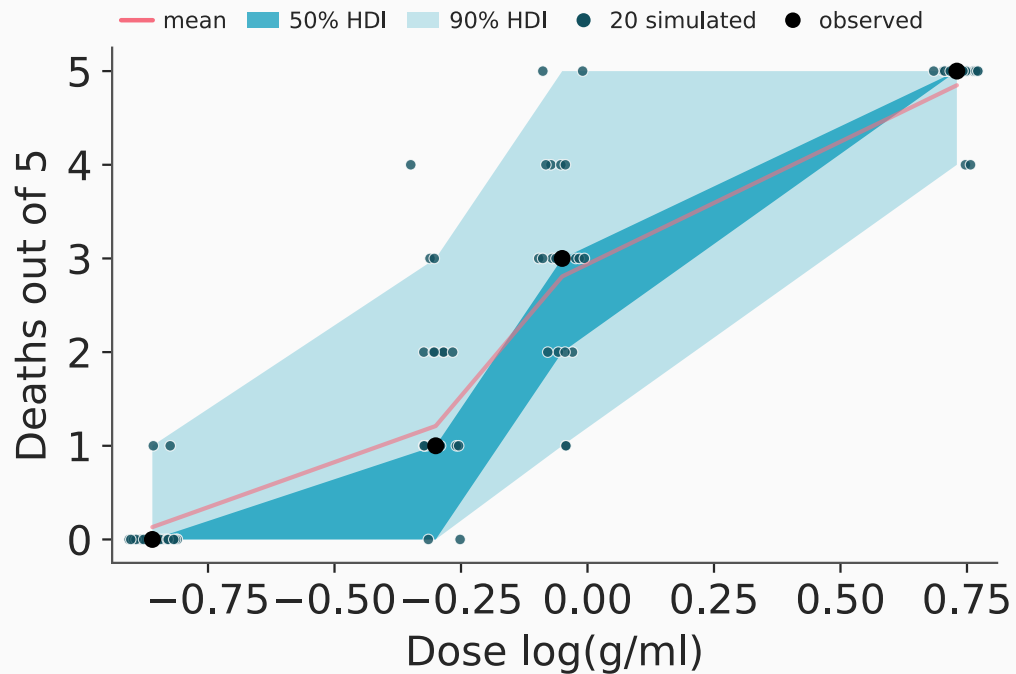
mean 930 mg/ml
90% HDI 716–1,119 mg/ml

LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.
- So every quantity comes with its own uncertainty.
- Here *how well* we know LD50 matters as much as its value.

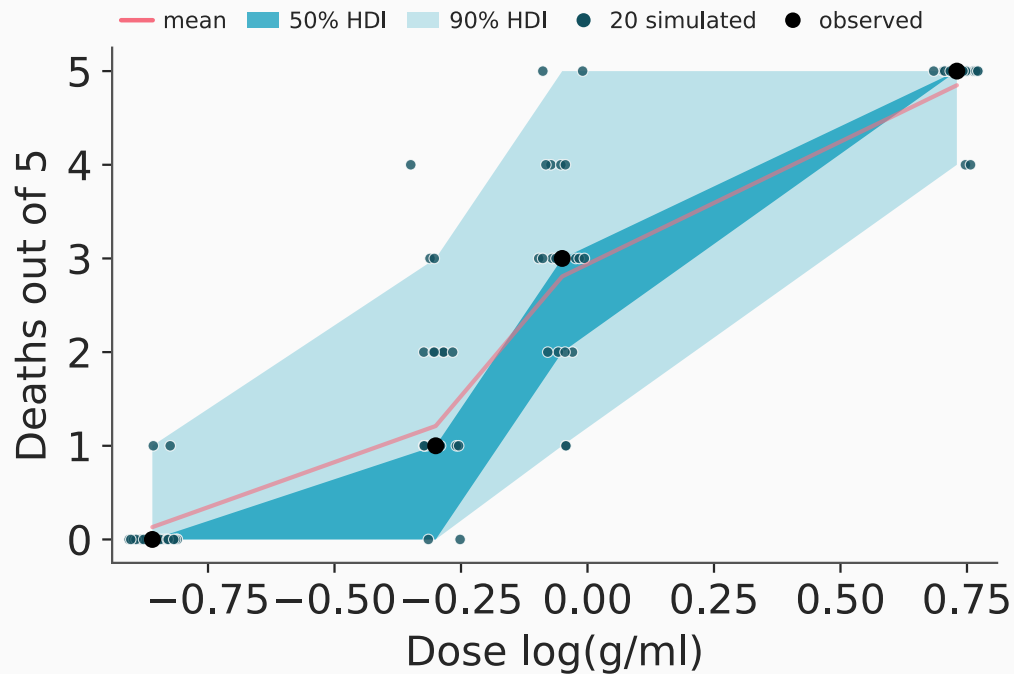


Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

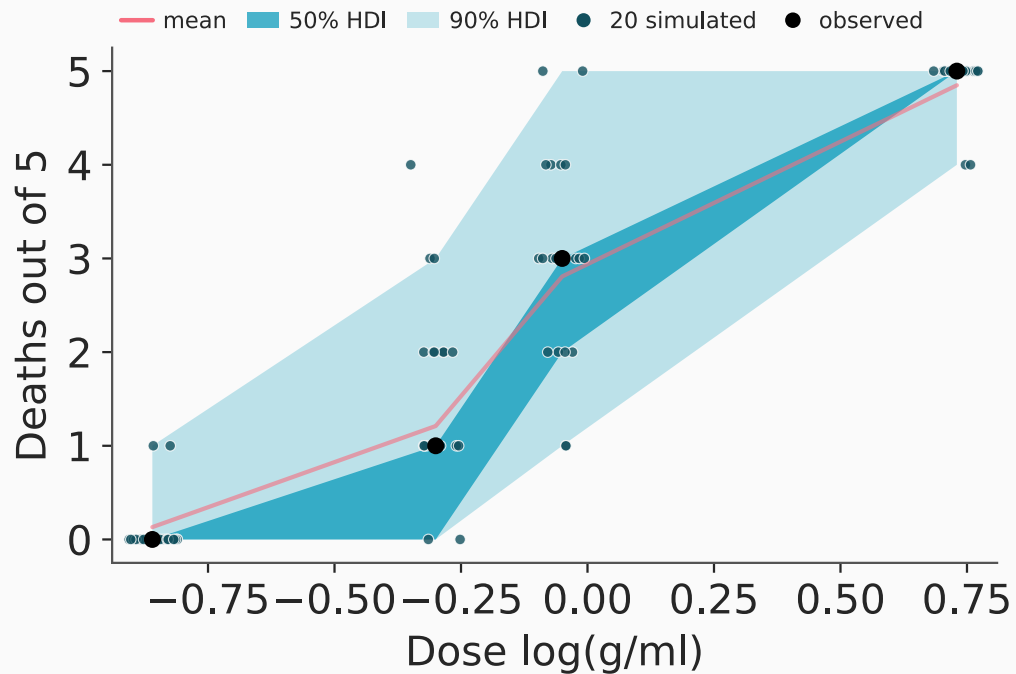
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- A workflow check meant to be done before looking at the posterior.

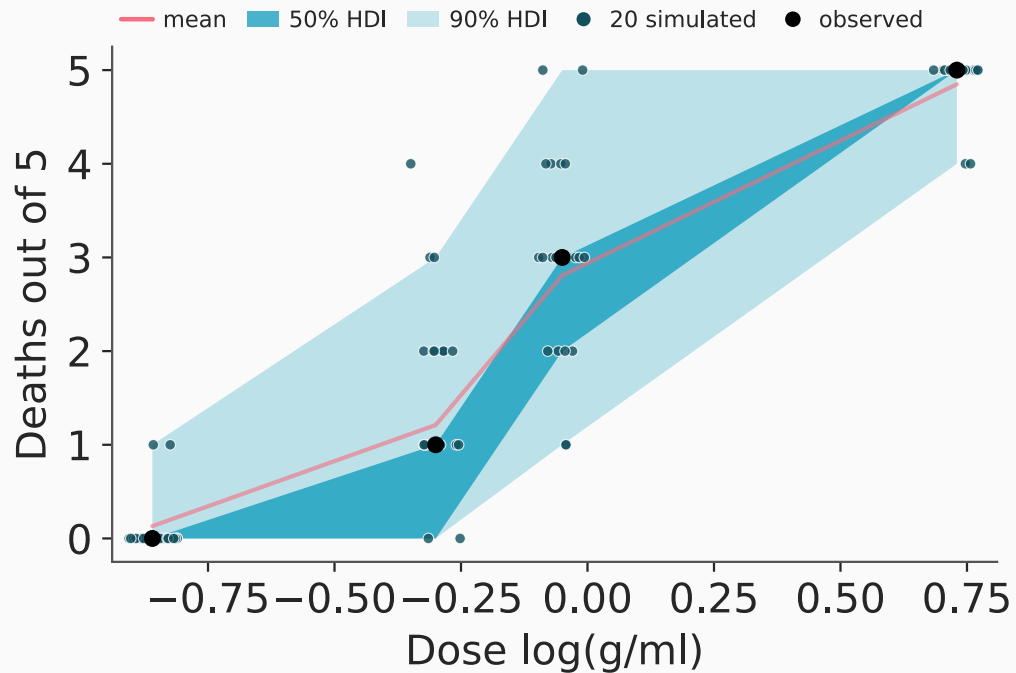
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- A workflow check meant to be done before looking at the posterior.
- Workflow is not lockstep: going back to check is always fine.

Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

- A workflow check meant to be done before looking at the posterior.
- Workflow is not lockstep: going back to check is always fine.
- We understand data better than we understand parameters.

The Bayesian toolkit

Four libraries, one workflow



Write the model, sample the posterior

pymc.io



Choose and check priors

preliz.readthedocs.io



Diagnose, summarize, and plot

python.arviz.org



Regression models from a formula

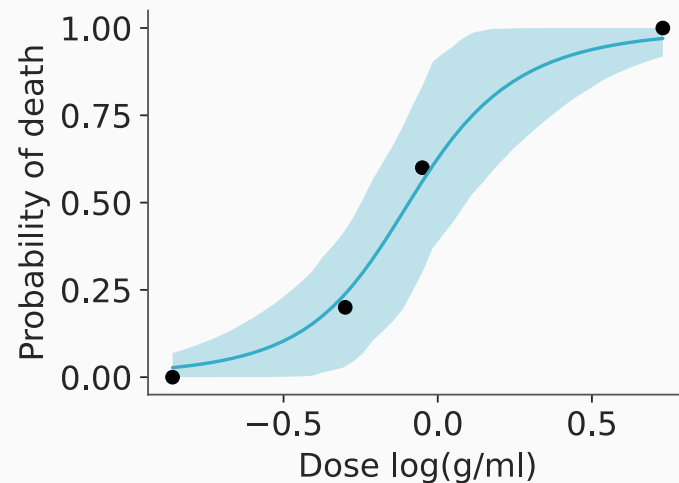
bambinos.github.io/bambi

Bambi: the same model as one formula

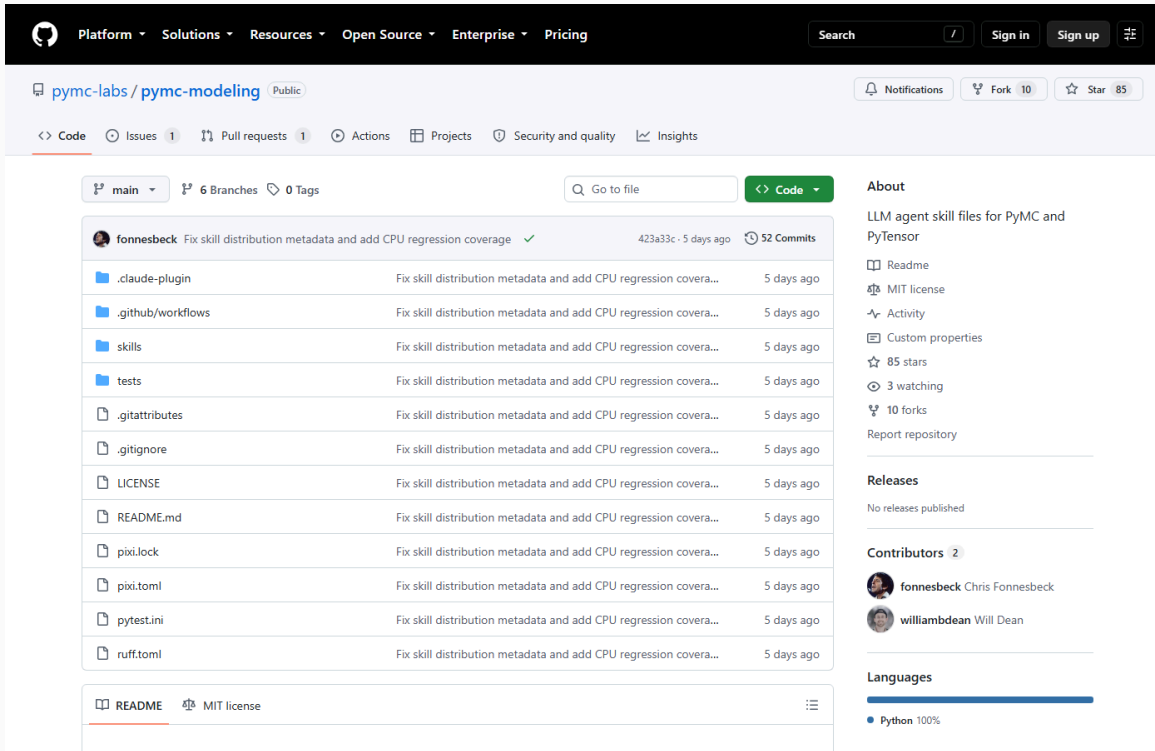
```
priors = {  
    "Intercept": bmb.Prior("Normal", mu=0, sigma=5),  
    "dose": bmb.Prior("HalfNormal", sigma=5),  
}  
model = bmb.Model("p(deaths, n) ~ dose", data,  
                  family="binomial", priors=priors)  
idata = model.fit()  
bmb.interpret.plot_predictions(model, idata, "dose")
```

Bambi: the same model as one formula

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    "Intercept": bmb.Prior("Normal", mu=0, sigma=5),  
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Coding assistants can learn current PyMC practice

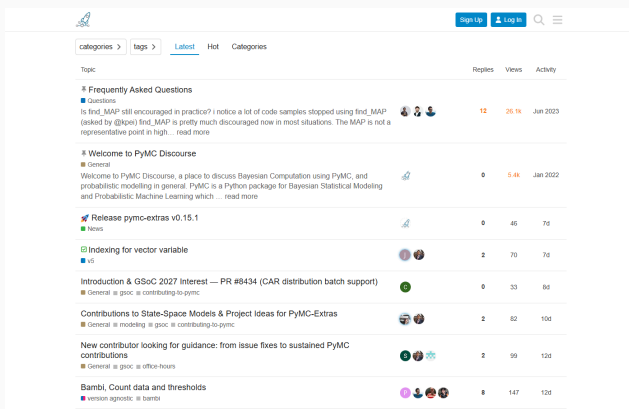


github.com/pymc-labs/pymc-modeling

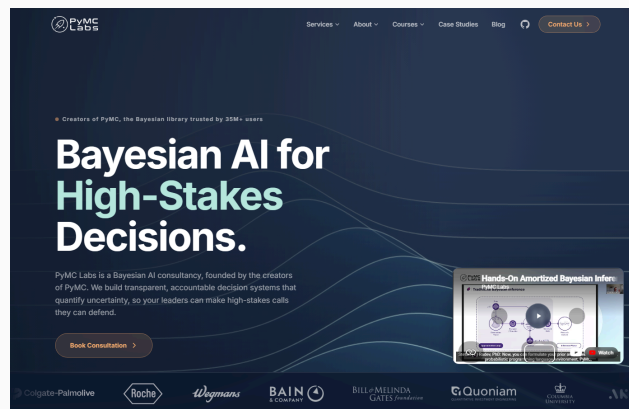
Skill files from PyMC Labs that teach AI coding assistants current PyMC and ArviZ practice.

We used them to build this course's notebooks.

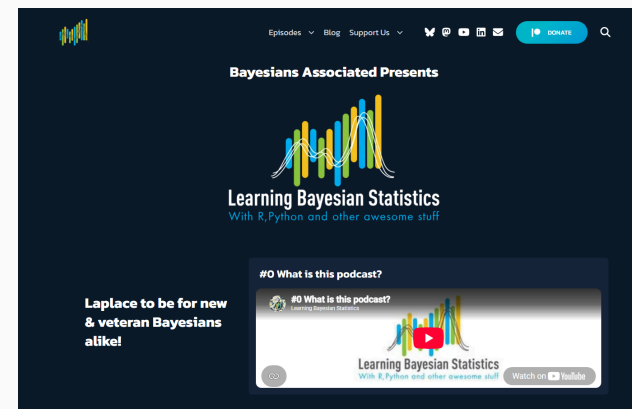
Where to ask, hire, and listen



Ask: PyMC Discourse
discourse.pymc.io



Hire: PyMC Labs
pymc-labs.com



Listen: Learning Bayesian Statistics
learnbayesstats.com

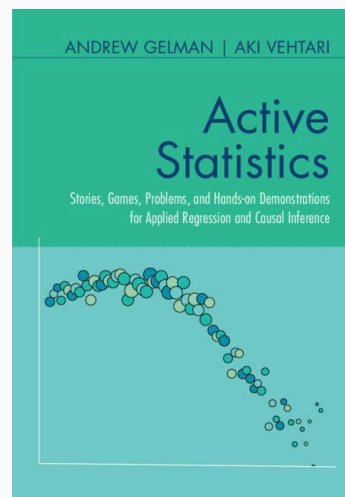
Books worth reading next



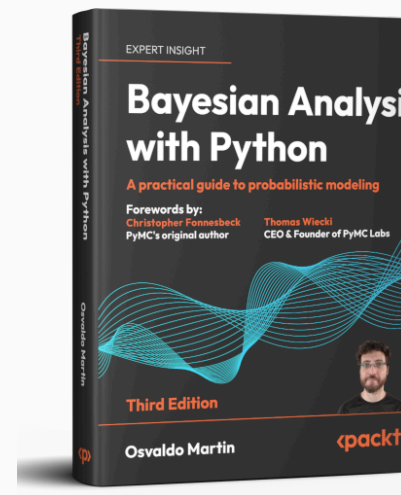
McElreath
Statistical Rethinking



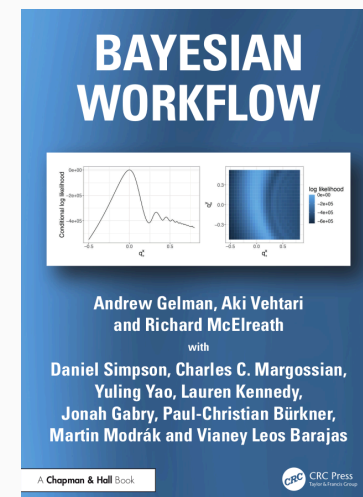
Gelman, Hill & Vehtari
Regression and Other Stories



Gelman & Vehtari
Active Statistics



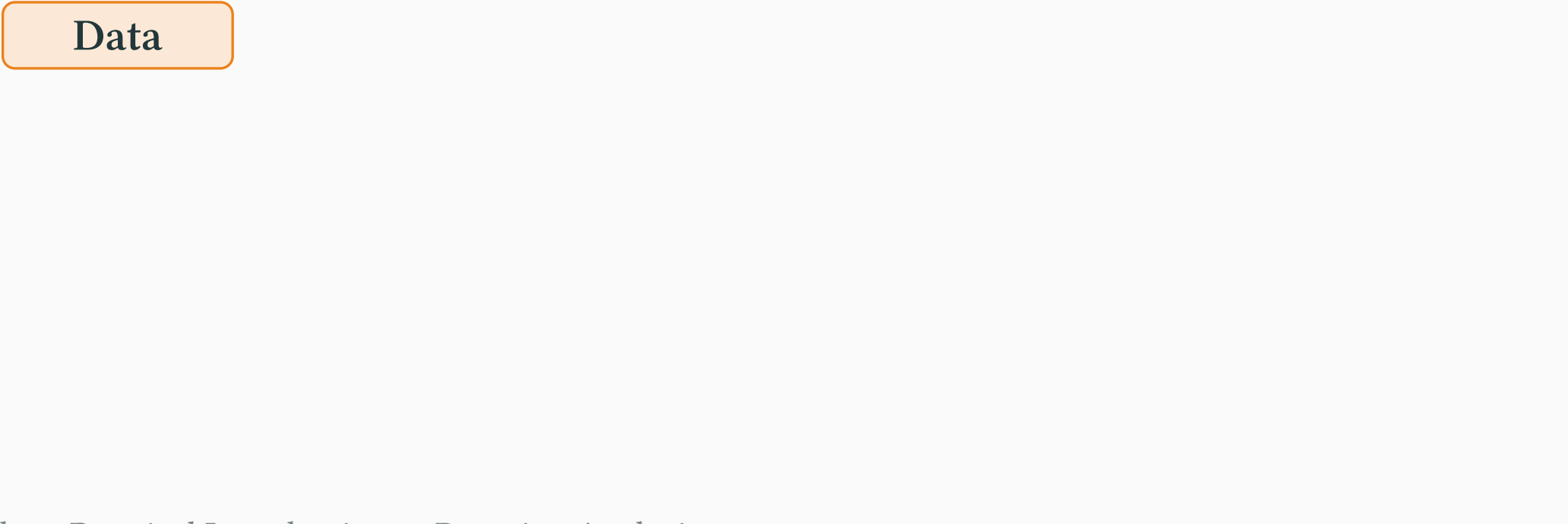
Martin
Bayesian Analysis with Python



Gelman, Vehtari et al.
Bayesian Workflow

The Bayesian workflow

The workflow starts with a four-step loop



Data

The workflow starts with a four-step loop



The workflow starts with a four-step loop



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The workflow starts with a four-step loop



Even at this level, we allow for

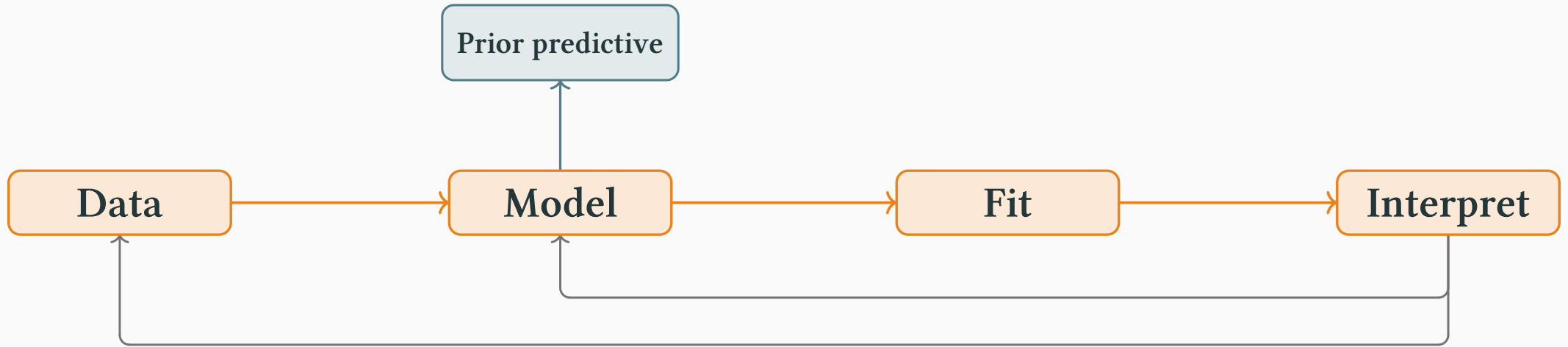
different interpretations · iterative experimental design

Checks make each transition accountable



Validation belongs at each step

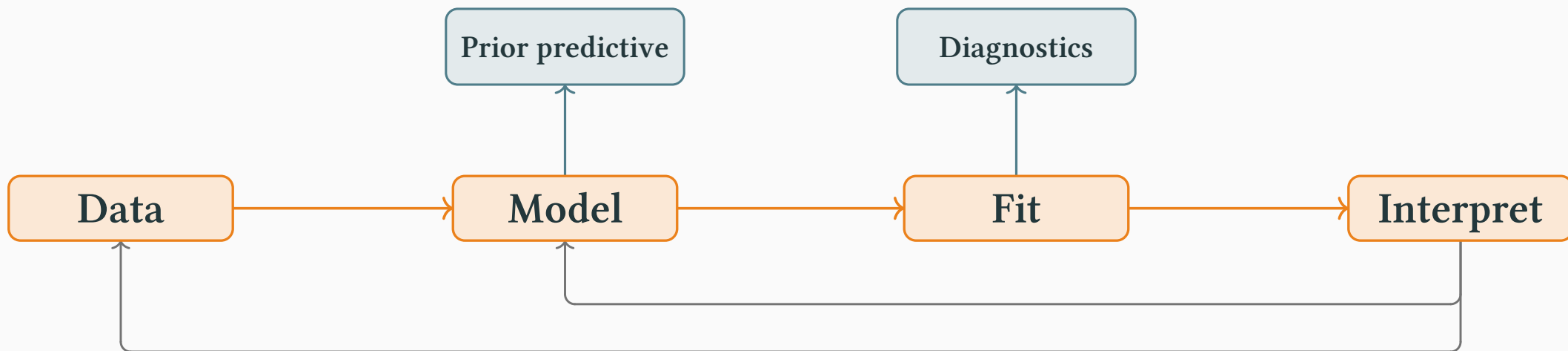
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Prior predictive

tests model coherence

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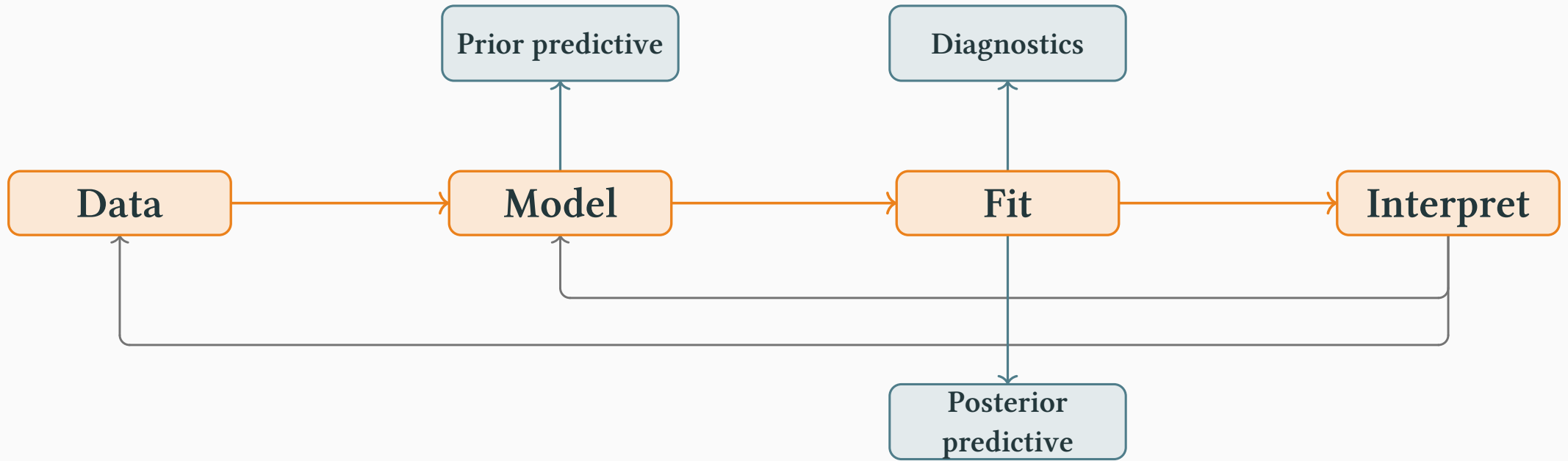
Prior predictive

Diagnostics

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test whether fitting converged

Checks make each transition accountable



Validation belongs at each step

Prior predictive

Diagnostics

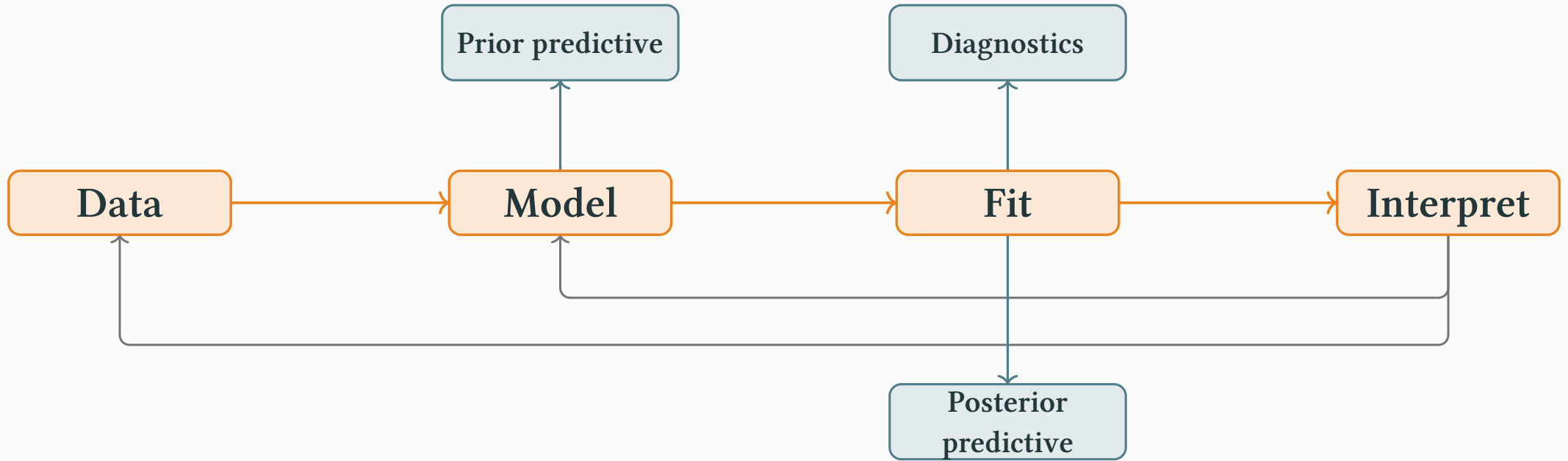
Posterior predictive

tests model coherence

test whether fitting converged

tests fit to the observed data

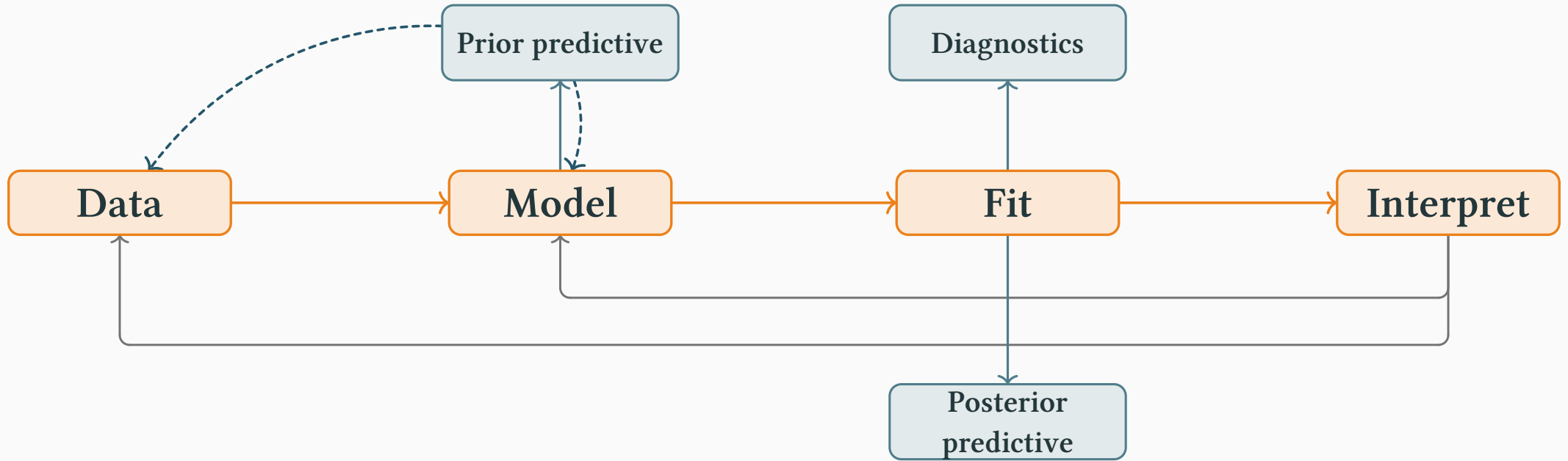
A failed check tells us where to return



A failed check

means something must change

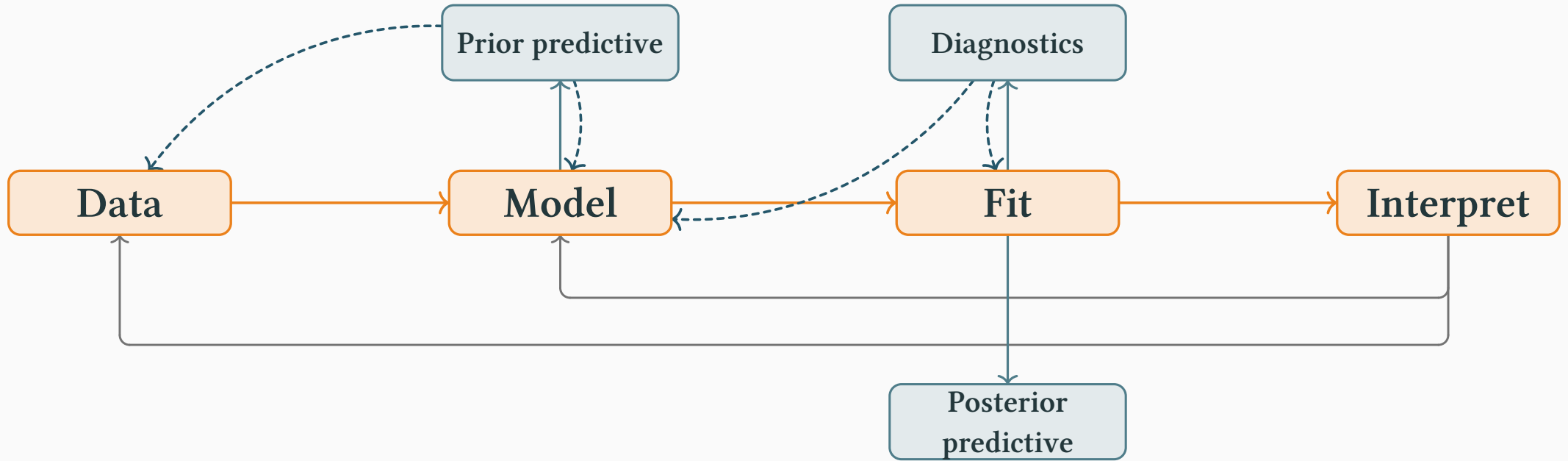
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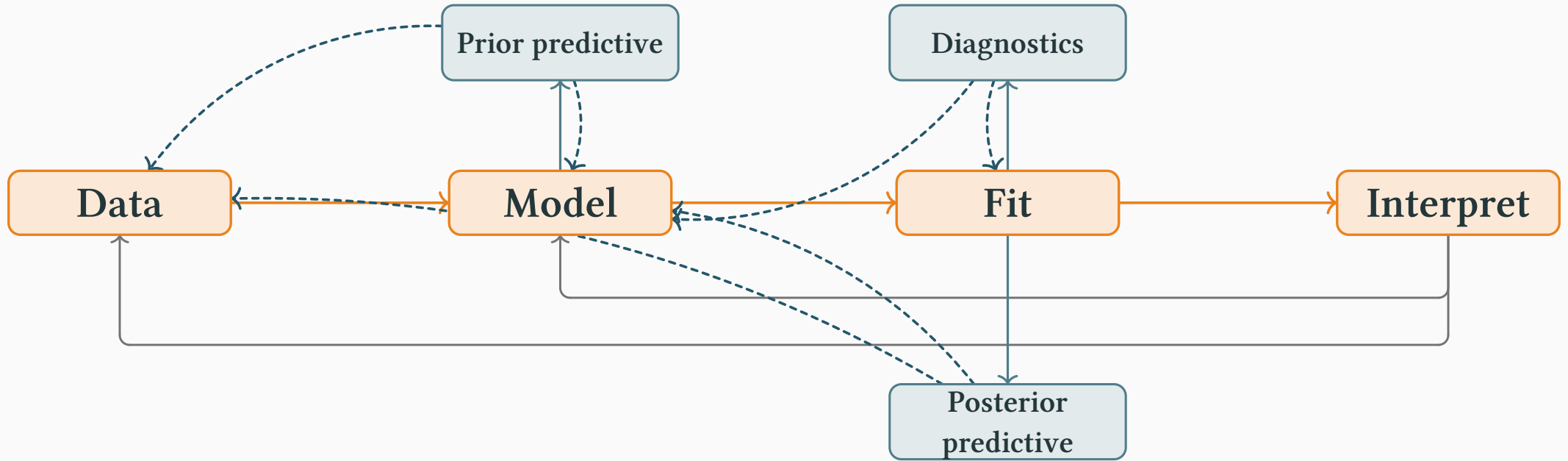
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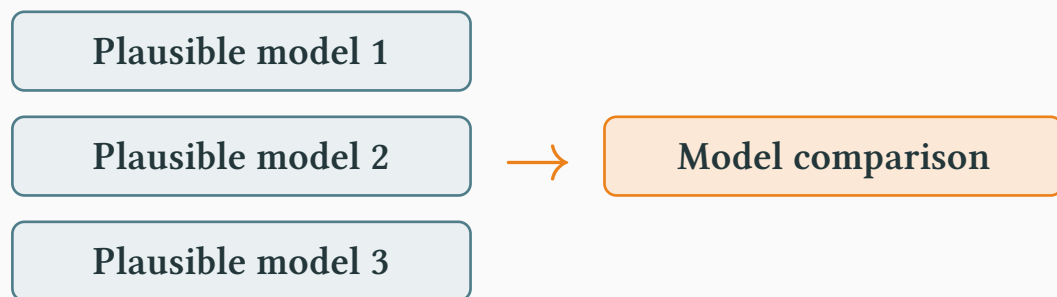
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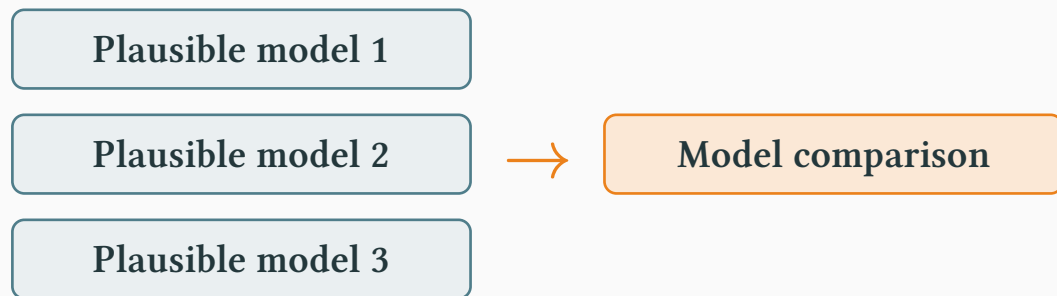
means something must change
revisit assumptions about the model or data
revisit the model or the computation
reassess how the model represents the data

Model comparison asks which useful story predicts best



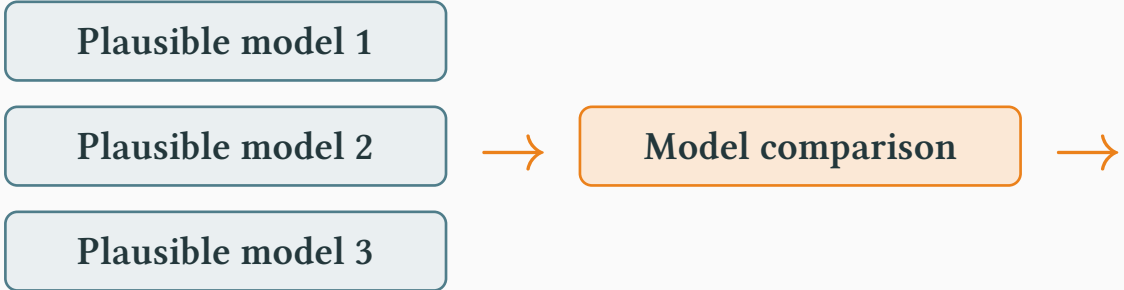
The workflow can lead to several **plausible models** with different interpretations.

Model comparison asks which useful story predicts best



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Model comparison asks which useful story predicts best

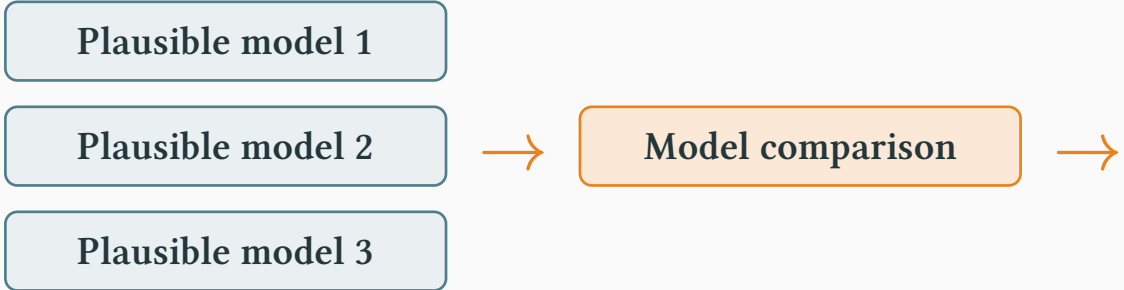


Leave-one-out cross-validation			
Model	LOO	SE	Δ LOO
Model 1	-124.8	4.5	13.0
Model 2	-112.4	0.3	0.6
Model 3	-111.8	-	0.0

illustrative values; larger LOO is better

The workflow can lead to several **plausible models** with different interpretations.
It is also a useful way to think about frequentist “null models.”
Model comparison is a basic tool in Bayesian data analysis.

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The workflow can lead to several **plausible models** with different interpretations.
It is also a useful way to think about frequentist “null models.”
Model comparison is a basic tool in Bayesian data analysis.
All models are wrong; some models are informative.

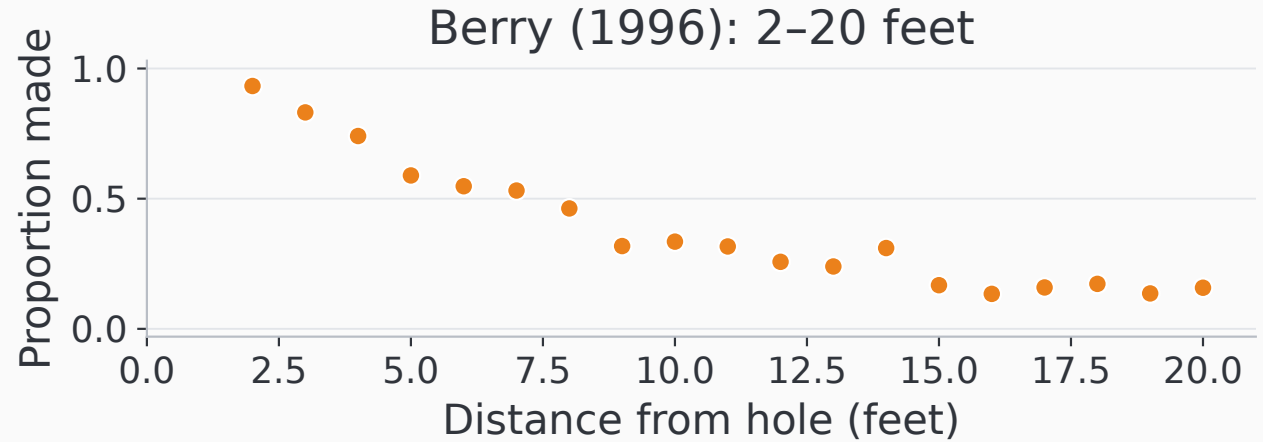
Working through the workflow: golf putting

The data



AI-generated course illustration

Berry (1996): professional putting summaries at rounded distances from 2 to 20 feet.



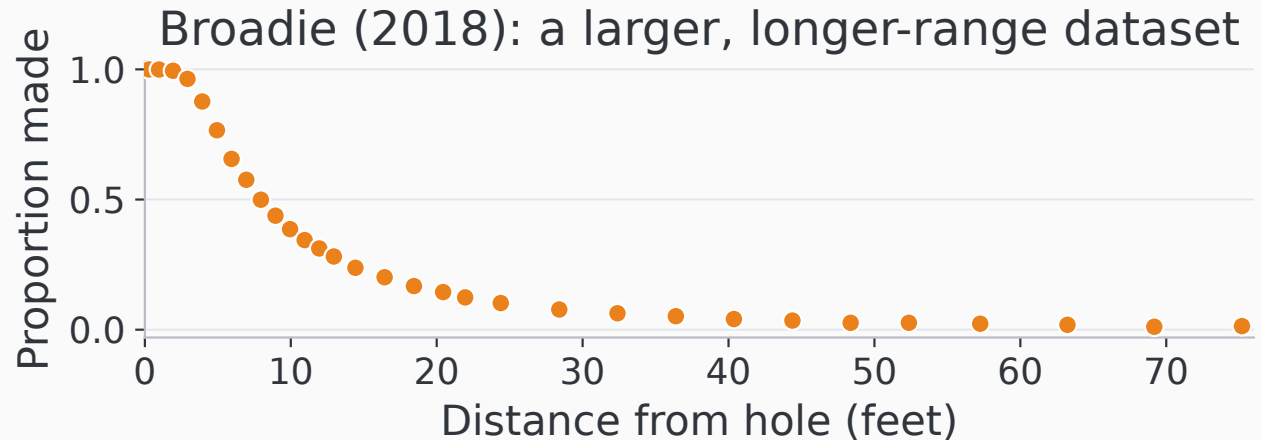
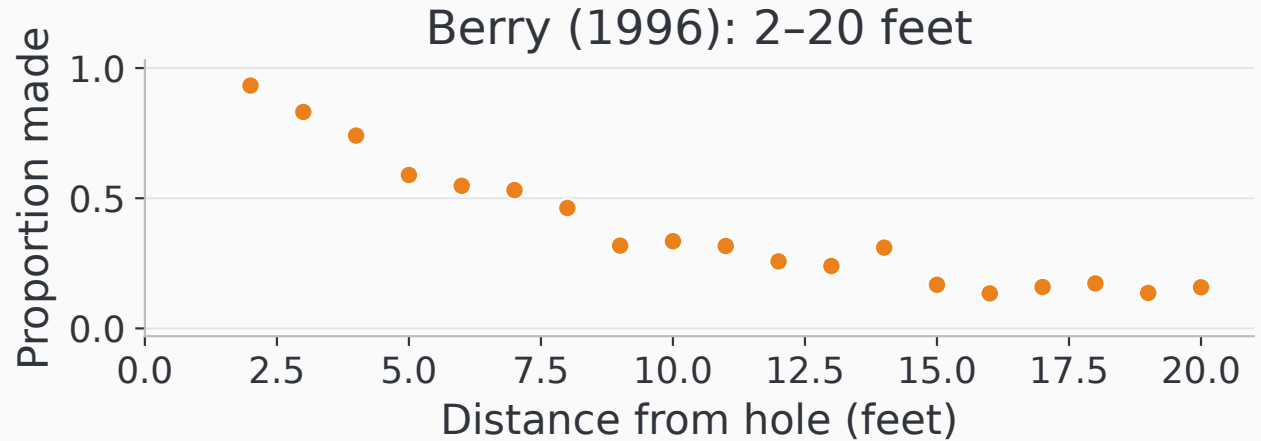
The data



AI-generated course illustration

Berry (1996): professional putting summaries at rounded distances from 2 to 20 feet.

Broadie (2018): a later, much larger dataset extending to 75 feet.



Starting with a simple model is easy and informative

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$$\text{logit}(p_j) = \alpha + \beta x_j$$

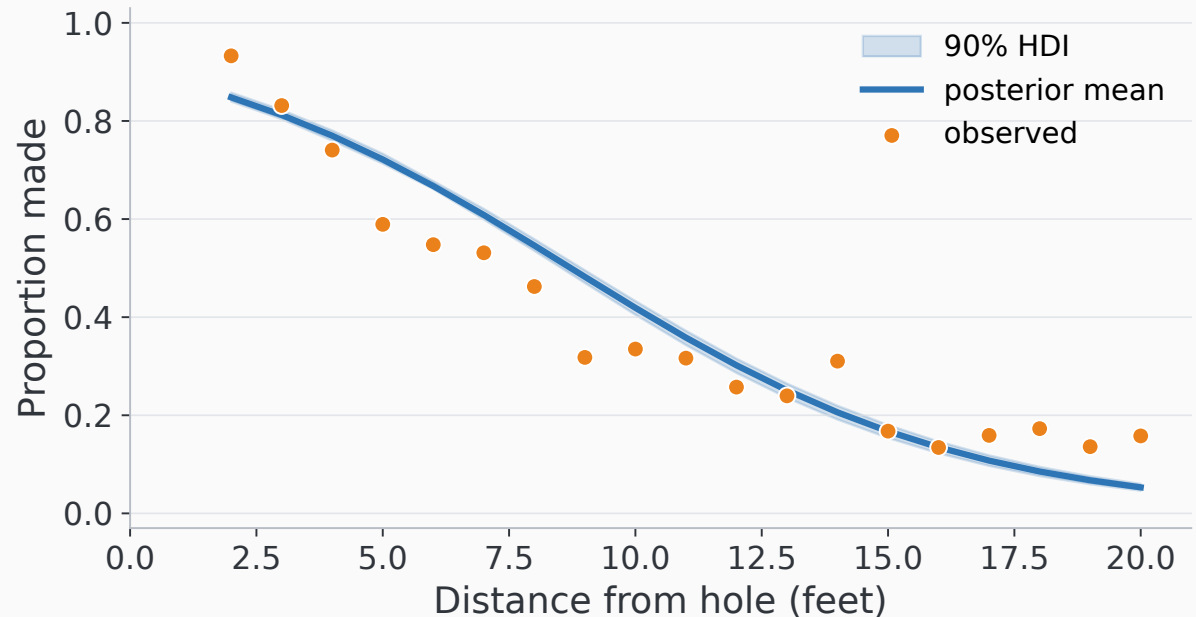
```
with pm.Model() as model:
    alpha = pm.Normal("alpha", 0, 3)
    beta = pm.Normal("beta", 0, 0.5)
    logit_p = alpha + beta *
distance
    pm.Binomial(
        "made", n=attempts,
        logit_p=logit_p,
        observed=made,
    )
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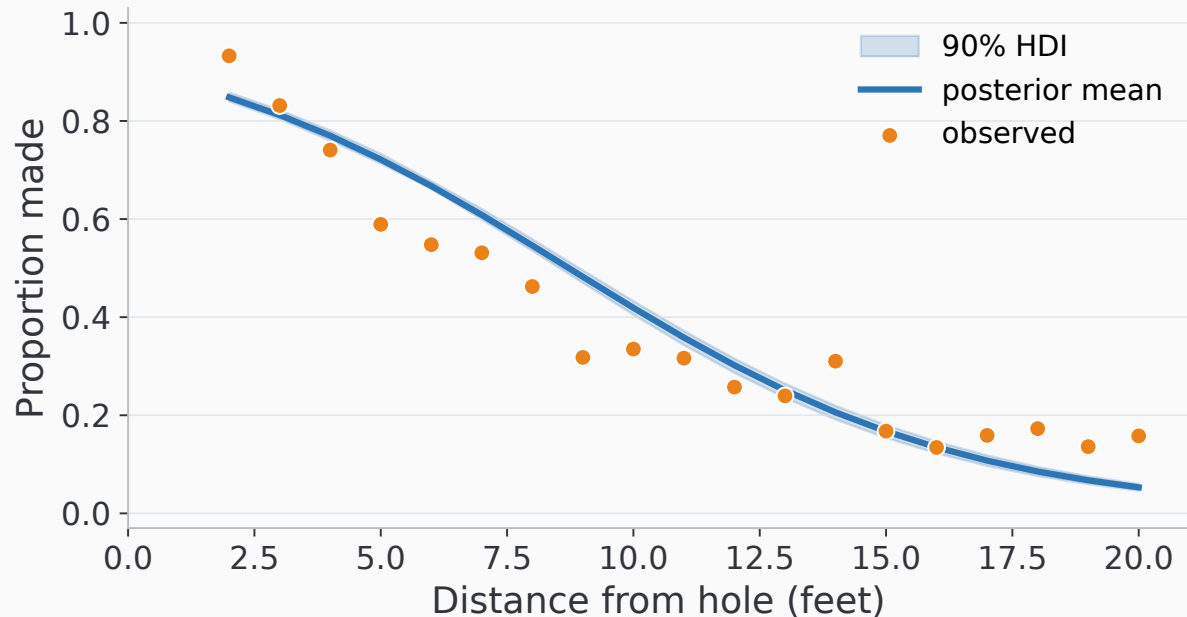


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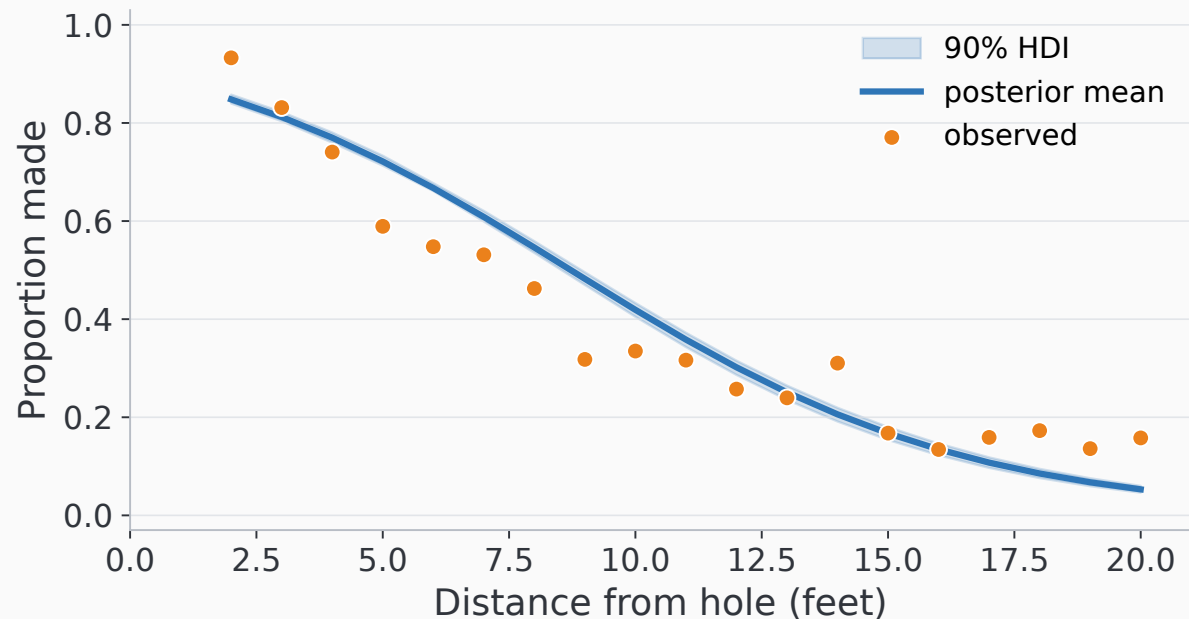
Prior and posterior predictive checks develop intuition for what it can generate.

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Prior and posterior predictive checks develop intuition for what it can generate.

What it does and does not capture tells us how much remains to explain.

Geometry turns distance into a success probability



$$\theta(x) = \arcsin\left(\frac{R-r}{x}\right)$$

$$p(x) = 2\Phi\left(\frac{\theta(x)}{\sigma}\right) - 1$$

Redrawn from Gelman et al., *Bayesian Workflow*, Fig. 25.3

Geometry turns distance into a success probability



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- A highly simplified model of putting physics

Geometry turns distance into a success probability



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- The only parameter is noise in the aiming angle, σ

Geometry turns distance into a success probability



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- These counts overwhelm almost any reasonable prior

Geometry turns distance into a success probability



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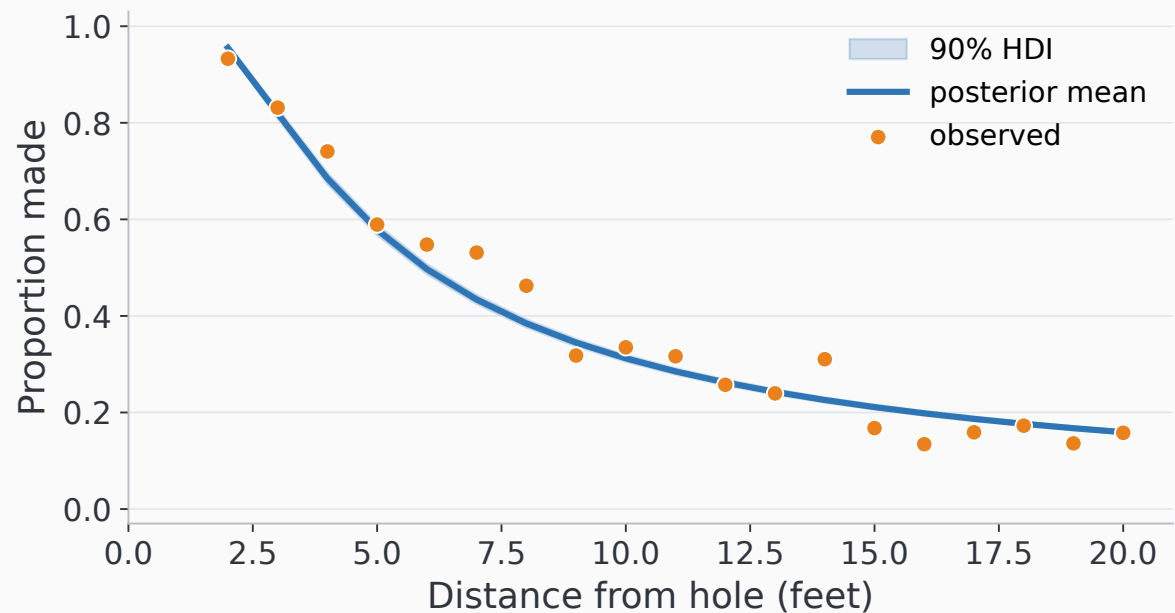
- A highly simplified model of putting physics
- The only parameter is noise in the aiming angle, σ
- These counts overwhelm almost any reasonable prior
- It is still healthy to ask what a reasonable prior implies

The aiming error model

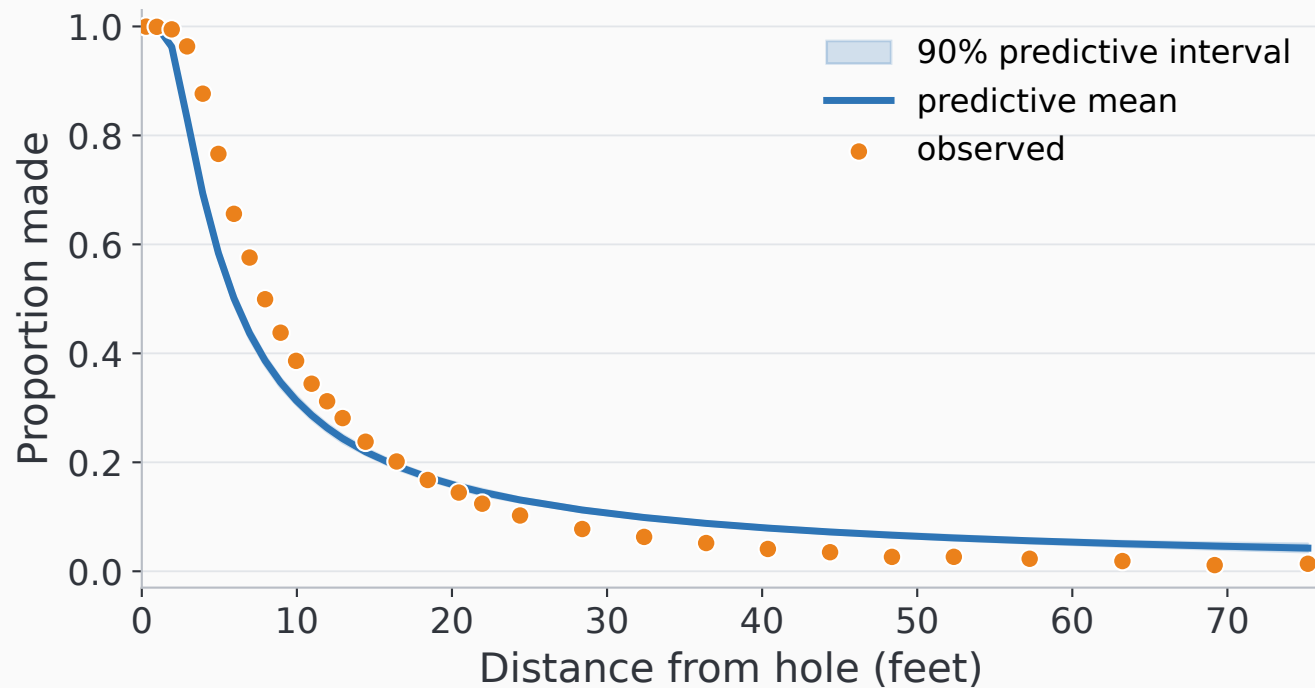
```
with pm.Model() as model:
    sigma_deg = pm.LogNormal(
        "sigma_deg", log(2), 0.7
    )
    theta = pm.math.arcsin(
        (cup_radius - ball_radius)
        / distance
    )
    p = 2 * pm.math.invprobit(
        theta / deg2rad(sigma_deg)
    ) - 1
    pm.Binomial(
        "made", n=attempts,
        p=p, observed=made,
    )
```

The aiming error model

```
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    theta = pm.math.arcsin(
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        / distance
    )
    p = 2 * pm.math.invprobit(
        theta / deg2rad(sigma_deg)
    ) - 1
    pm.Binomial(
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    )
```

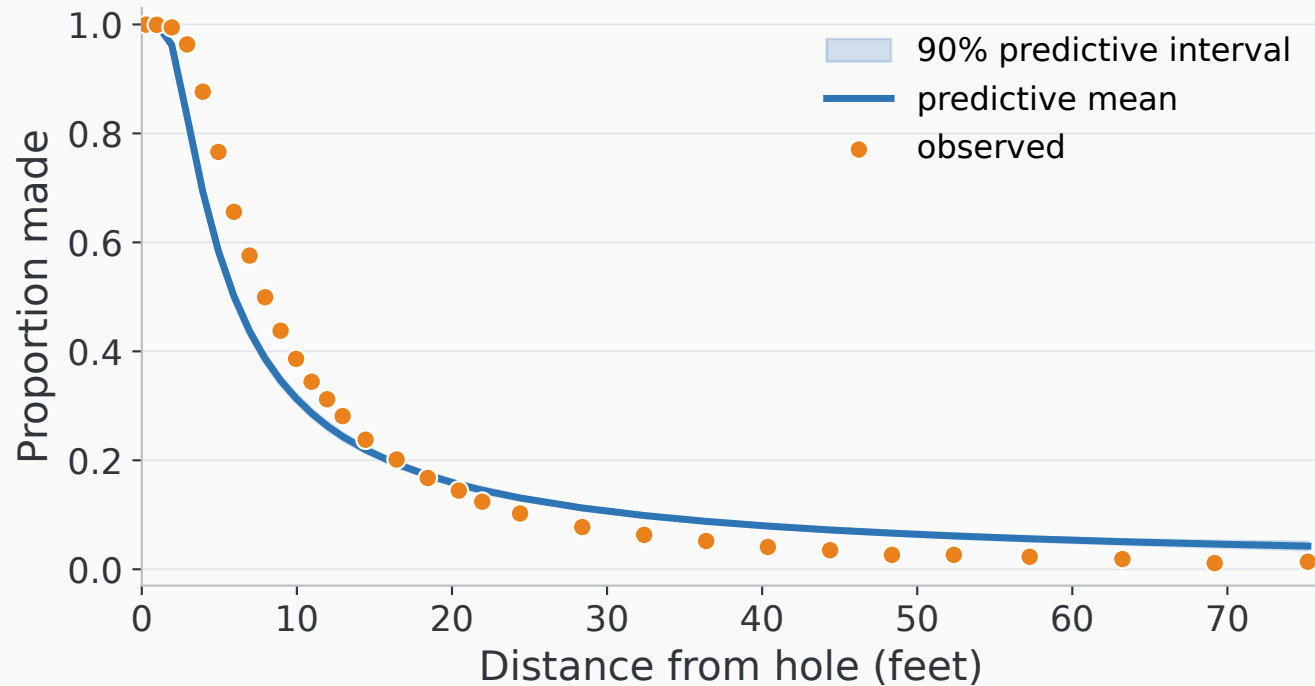


The aiming error model on new data



Data: Broadie (2018); comparison follows *Bayesian Workflow*, §25.3

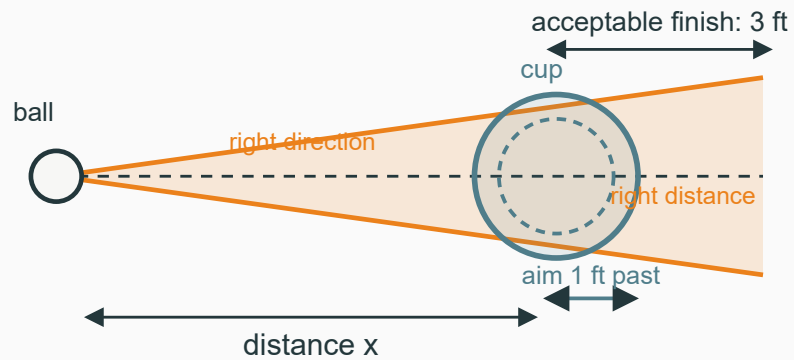
The aiming error model on new data



When we have faith in a model, we can test it against new data as those data become available.

Data: Broadie (2018); comparison follows *Bayesian Workflow*, §25.3

A new physical model

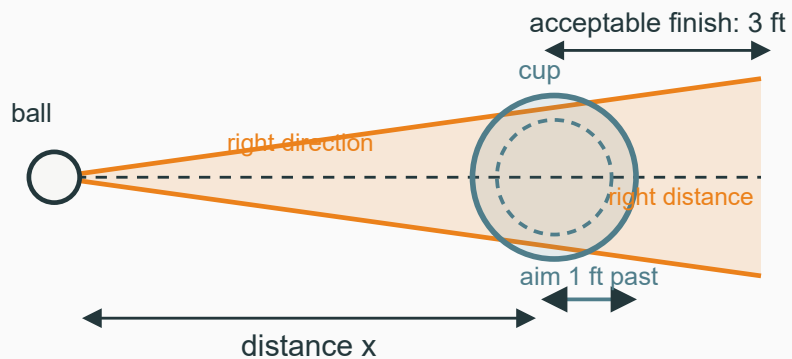


Redrawn from Gelman et al., *Bayesian Workflow*, Fig. 25.6

Direction

$$p_{\text{angle}(x)} = 2\Phi\left(\frac{\arcsin\left(\frac{R-r}{x}\right)}{\sigma_{\text{angle}}}\right) - 1$$

A new physical model



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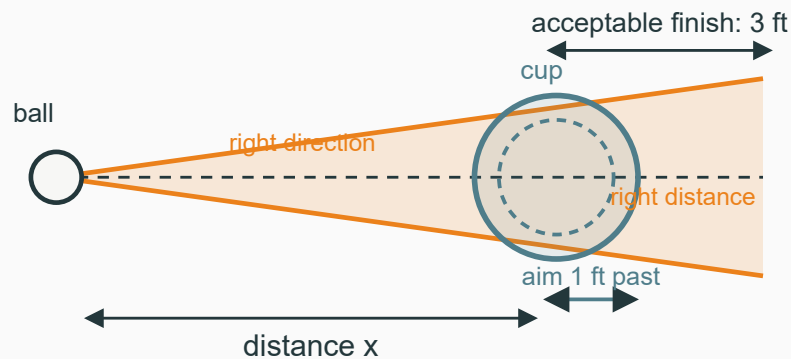
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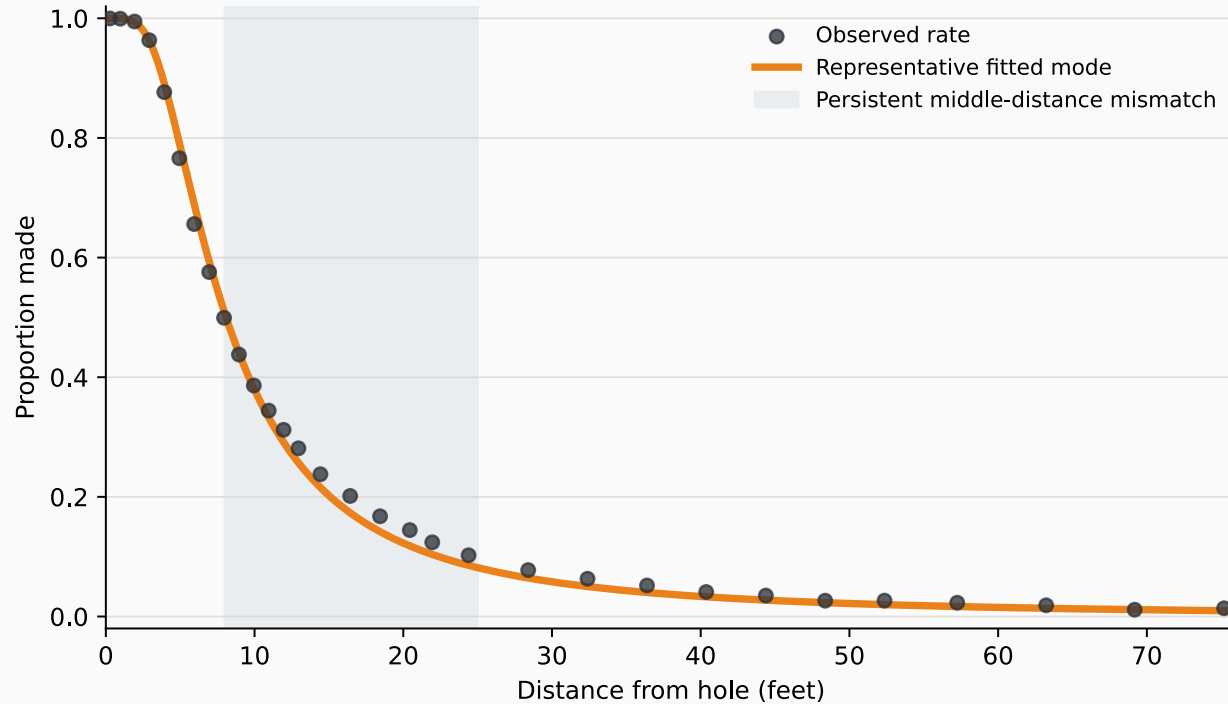
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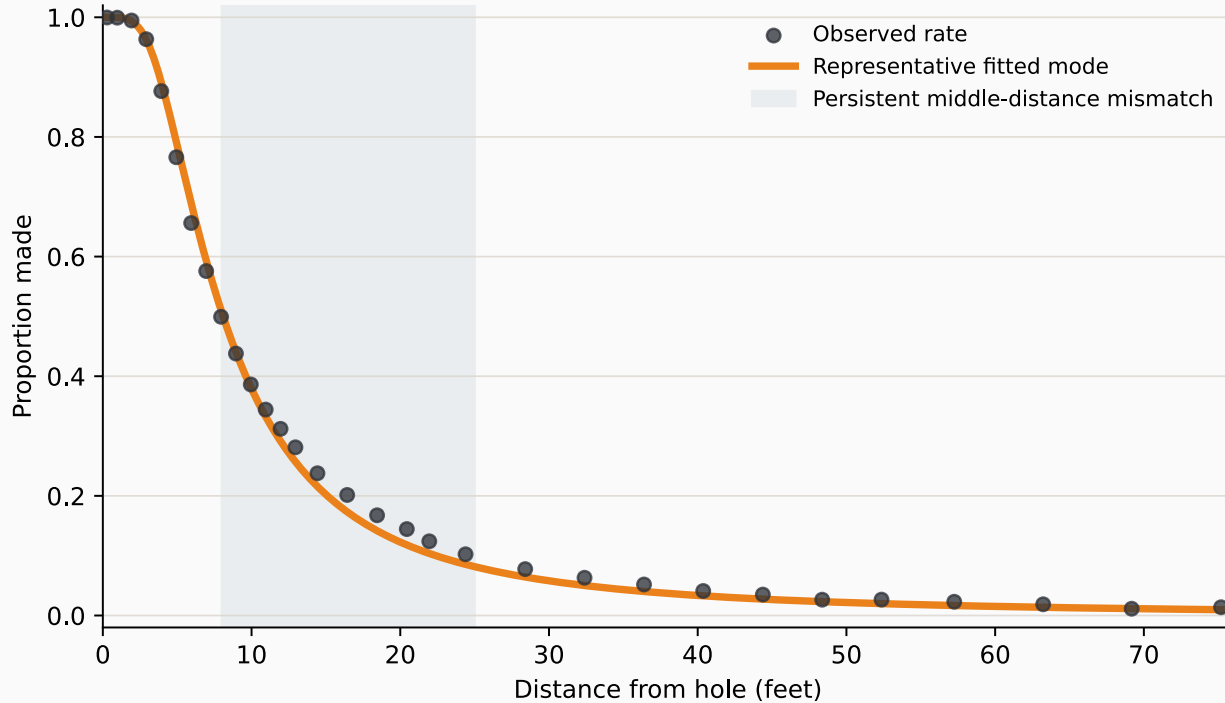
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The aiming + distance model

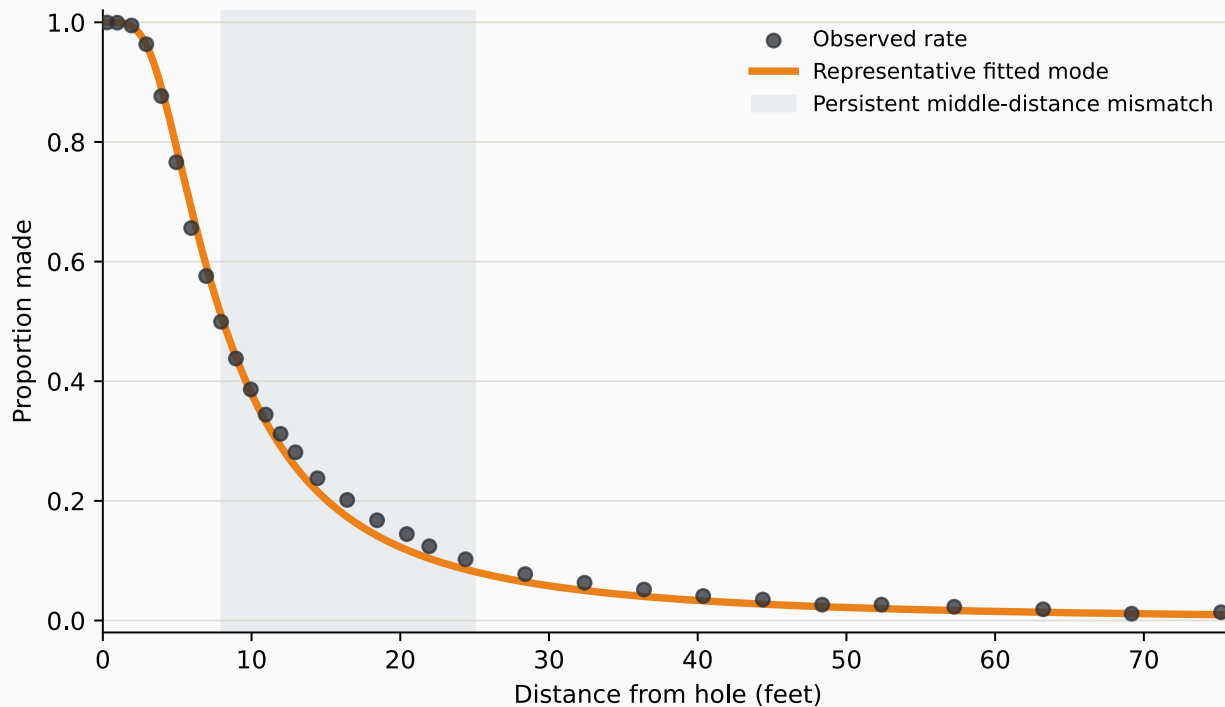


The aiming + distance model



An aiming and distance model improves fit and interpretability.

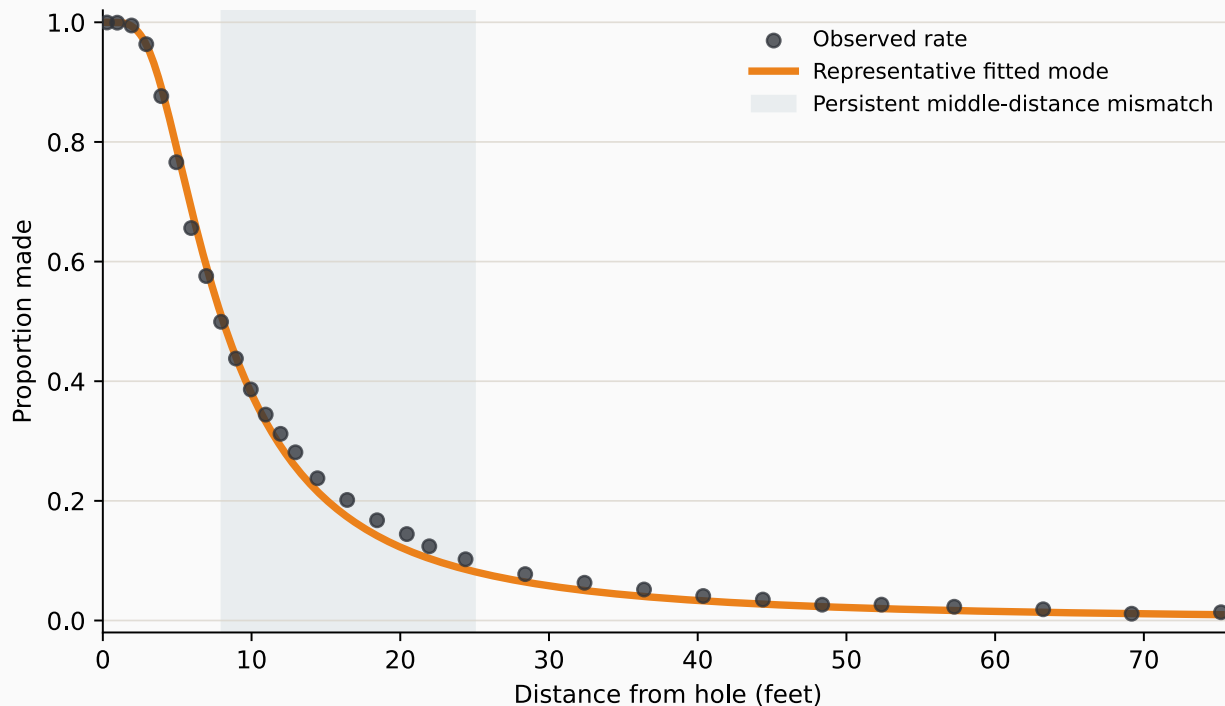
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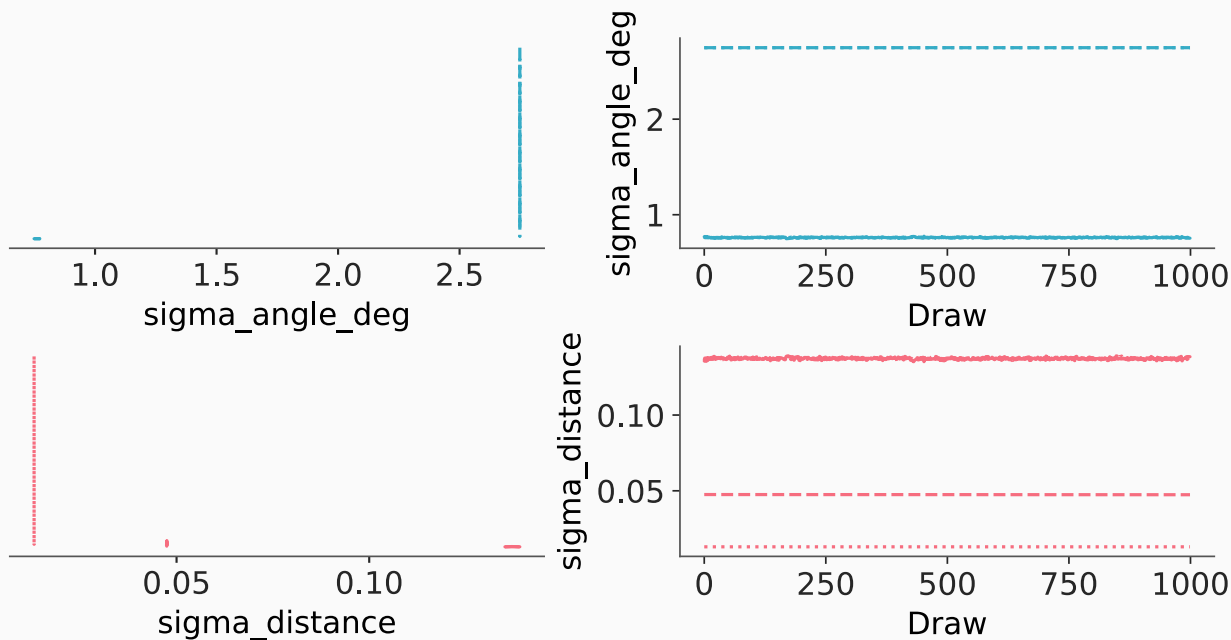


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A plausible curve is not yet a trustworthy posterior.

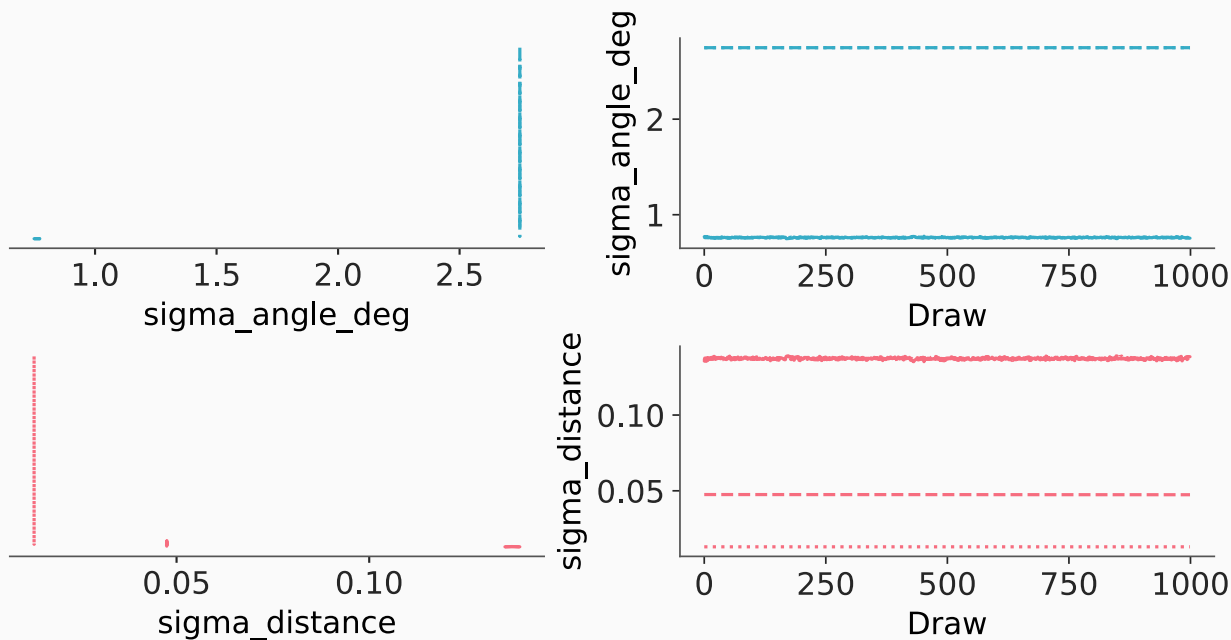
The aiming + distance model: have we sampled the posterior?



The chains disagree:

- $\hat{R} \approx 2$
- ESS ≈ 5 of 4,000 draws

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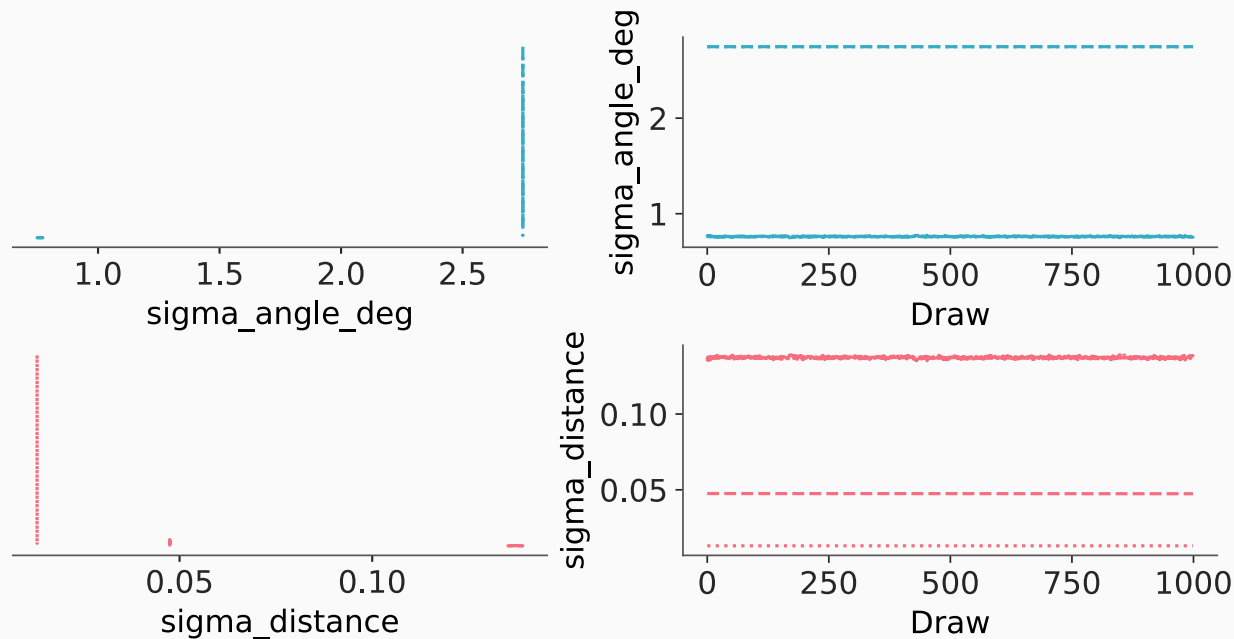


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We have not sampled the posterior.

So model fit is not even the question yet.

Each failed check suggests the next model—not the final model

- **Absorb local mismatch.** Huge short-putt counts dominate on the logit scale. Replace the Binomial with a Normal discrepancy model: **wrong, but better.**

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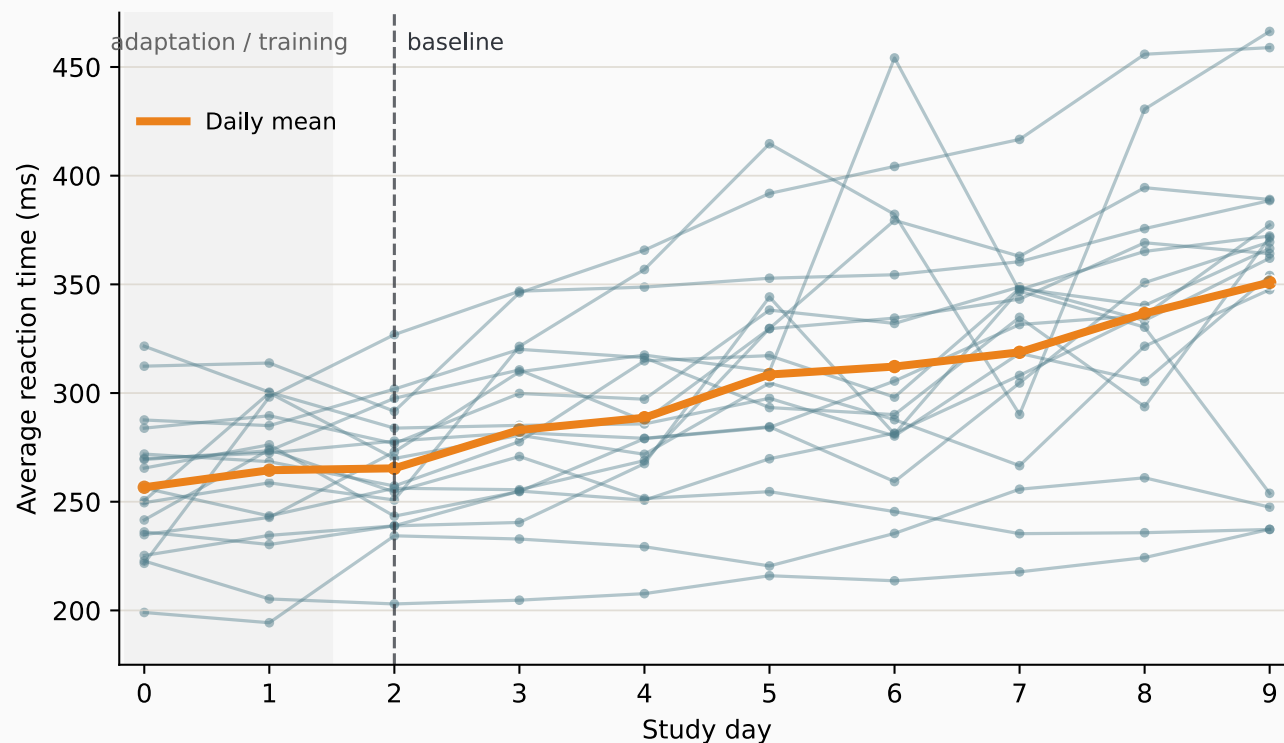
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- **Keep iterating critically.** Model building develops intuition—and the habit of asking what each apparent improvement actually explains.

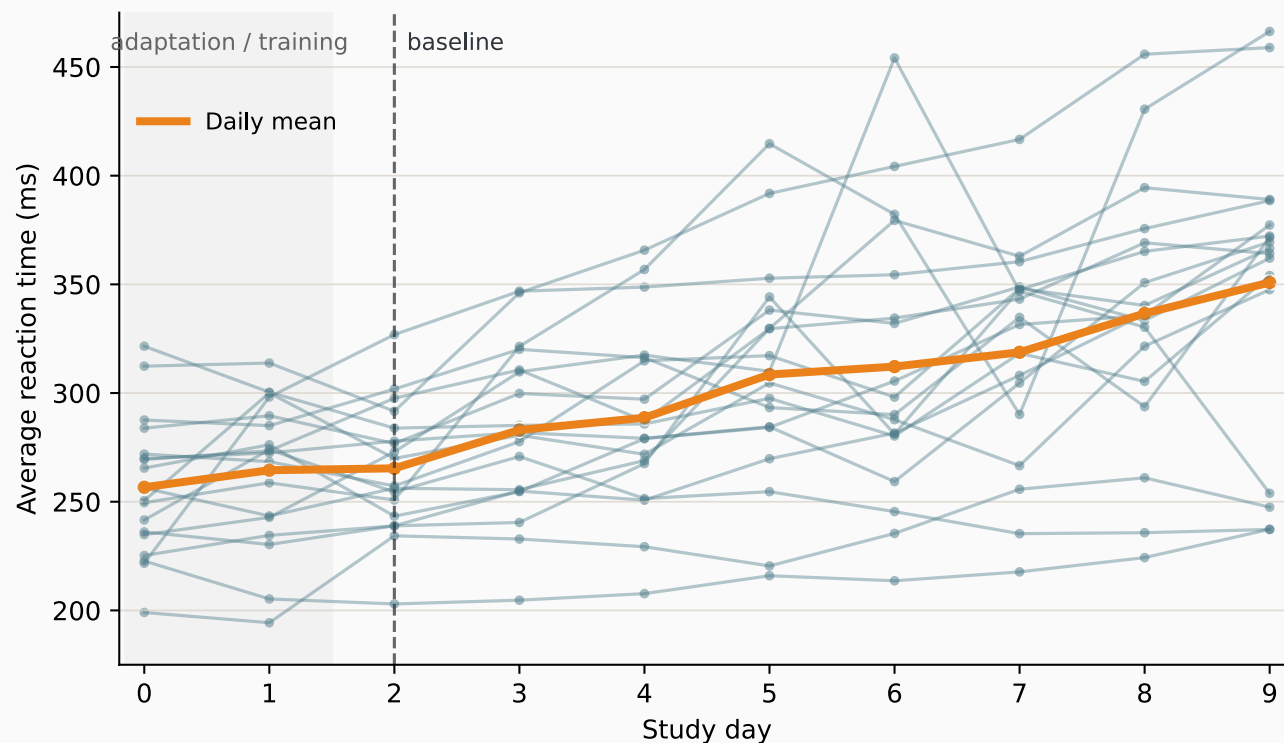
Becoming independent Bayesian analysts

The sleep deprivation data



Data: Belenky et al. (2003), distributed as `sleepstudy` in `lme4`

The sleep deprivation data

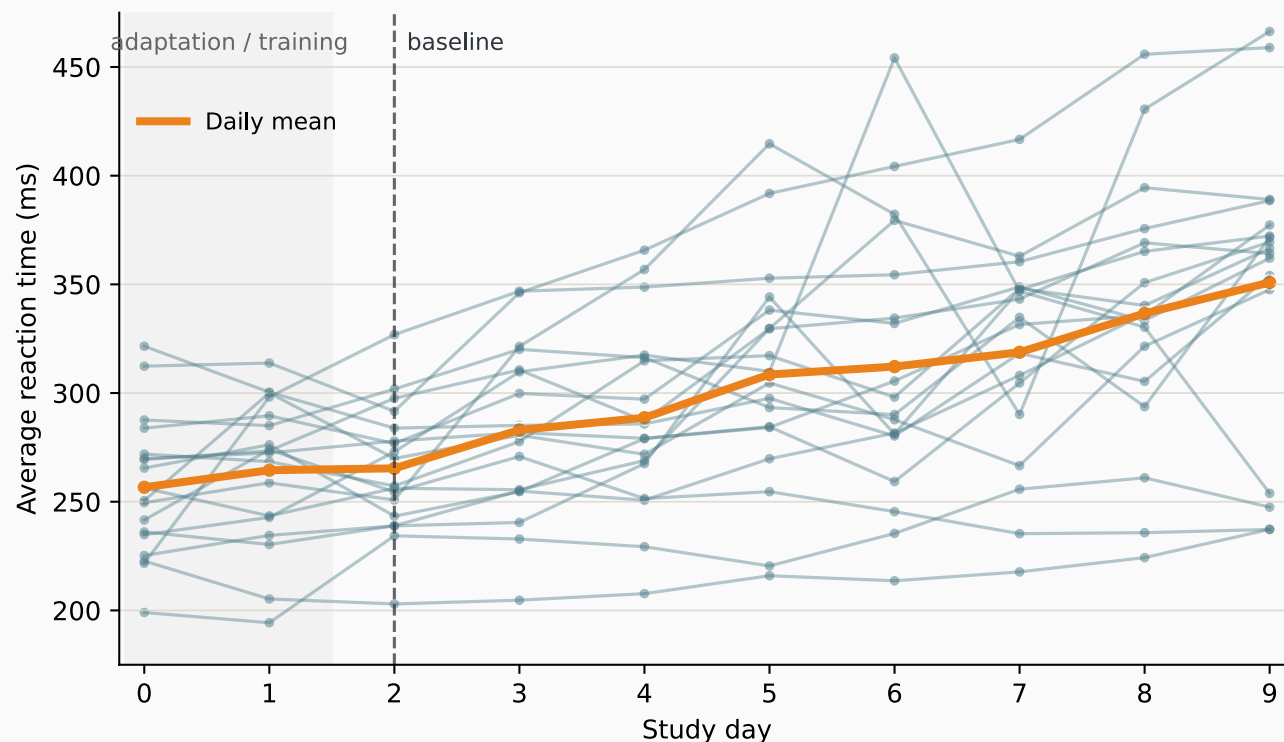


180 observations

- 18 participants
- 10 daily averages each

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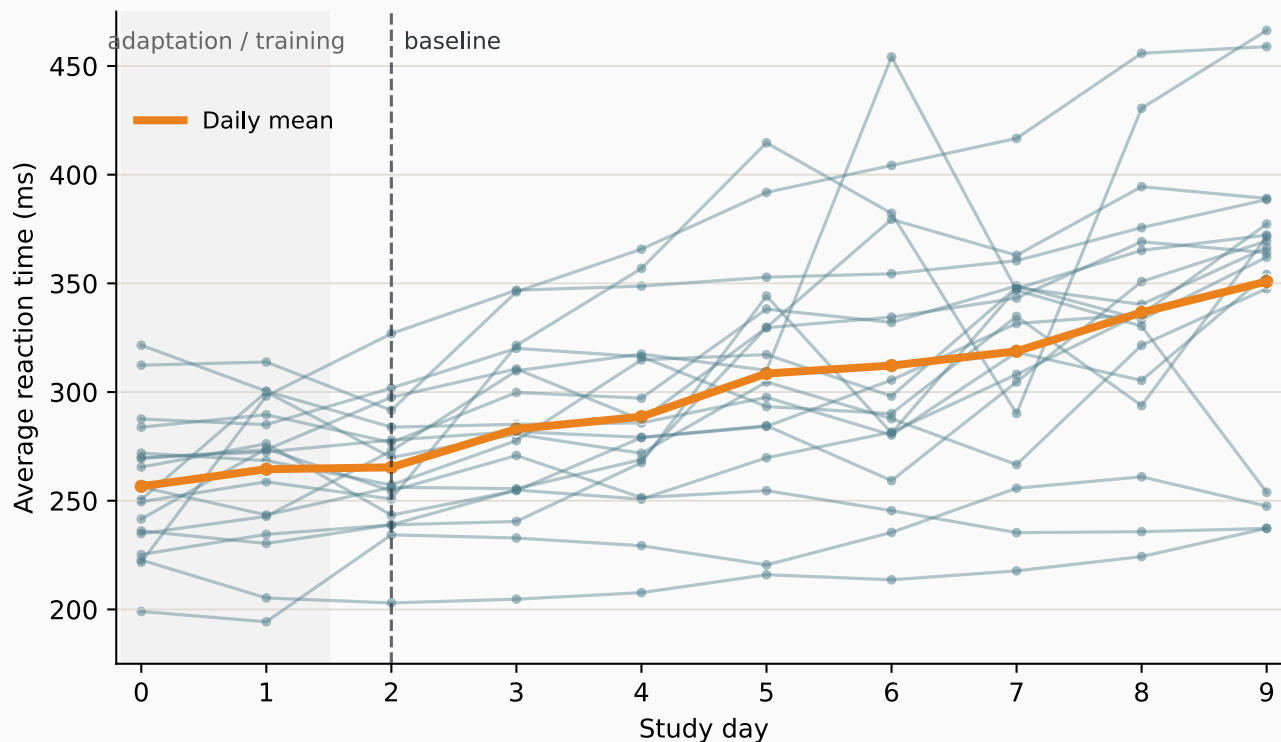
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The population trend is clear—yet participants differ in both baseline and change.

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Choose priors

- Use common sense
- Check prior predictions
- Handle practical constraints

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The goal is independence: formulate, fit, check, revise, and explain.